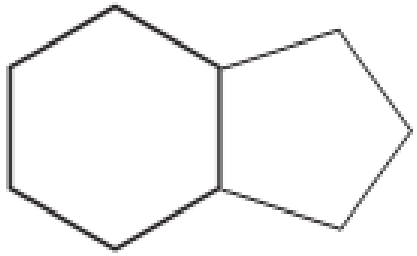
Stereochemistry Of Fused And Bridged Ring System

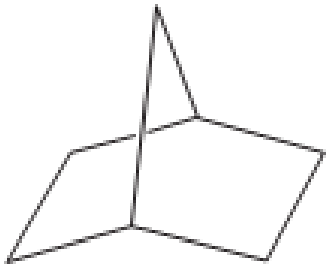
Bicyclic Compounds

Fused And Bridged Ring System

- Some cyclic compounds contain more than one ring.
- If two rings share two or more common atoms, the compound is called a **bicyclic compound**.
- If two rings have **a single** common atom, the compound is called a **spirocyclic compound**.

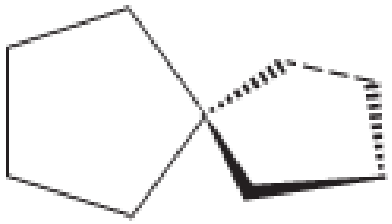


bicyclo[4.3.0]nonane



bicyclo[2.2.1]heptane

(bicyclic compounds)



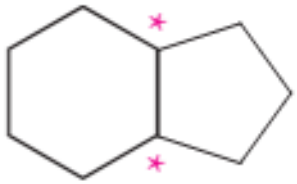
spiro[4.4]nonane

(a spirocyclic compound)

Fused And Bridged Ring System

- The **carbon atoms** at which two rings are joined in a bicyclic compound are called **bridgehead carbons.**

i. When the bridgehead carbons of a bicyclic compound are adjacent, the compound is classified as a **fused bicyclic compound:**



bridgehead carbons (*) at adjacent positions

a fused bicyclic compound

ii. When the bridgehead carbons are not adjacent, the compound is classified as a **bridged bicyclic compound:**



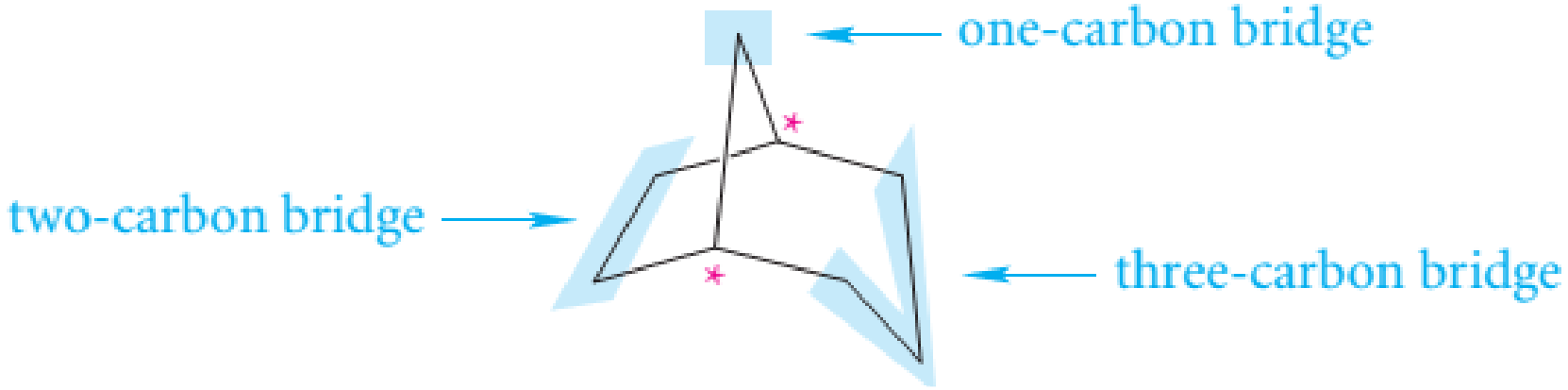
bridgehead carbons (*) not adjacent

a bridged bicyclic compound

Bicyclic Compounds

Fused And Bridged Ring System

- The nomenclature of bicyclic hydrocarbons is best illustrated by example:



- This compound is named as a bicyclo**octane** because it is a bicyclic compound containing a total of **eight** carbon atoms.
- The numbers in brackets represent the number of carbon atoms in the respective bridges, in order of decreasing size.

Bicyclic Compounds

Fused And Bridged Ring System

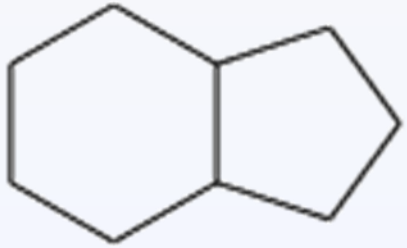
- Draw structures for each of the following compounds. Identify the bridgehead carbon atoms if they are present.

a. spiro[3.5]nonane

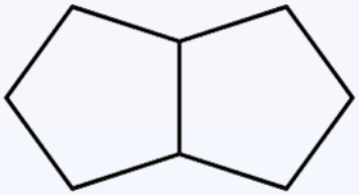
b. bicyclo[4.3.0]nonane

c. bicyclo[3.3.0]octane

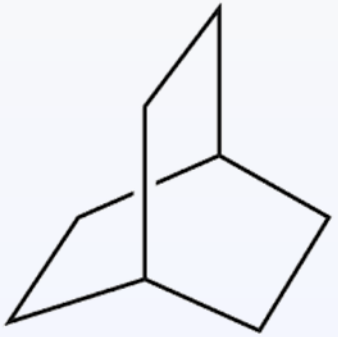
d. bicyclo[2.2.2]octane



bicyclo[4.3.0]nonane



bicyclo[3.3.0]octane

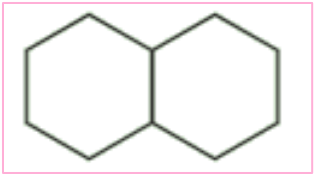


bicyclo[2.2.2]octane

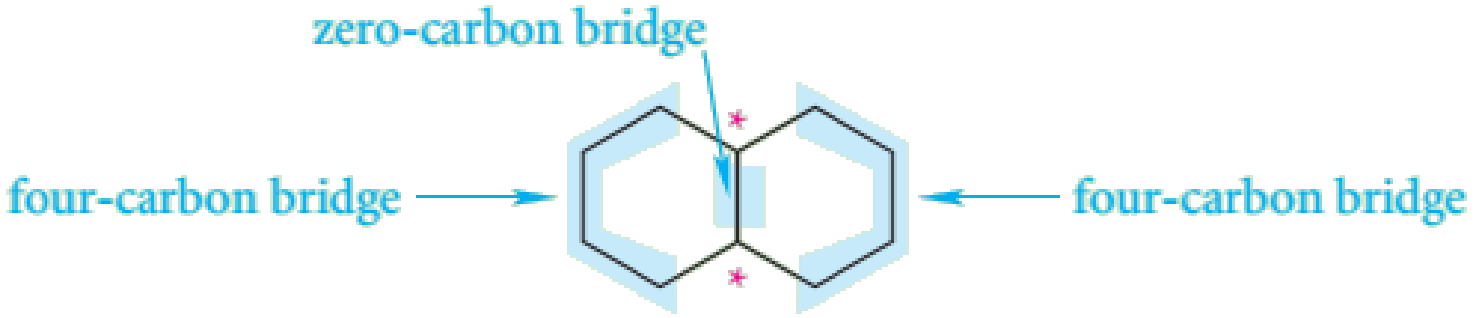
Bicyclic Compounds

Fused And Bridged Ring System

- Give the IUPAC name of the compound:
  (Its common name is decalin.)

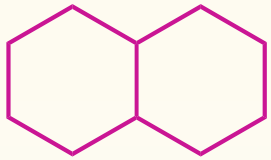


- The compound has two fused rings that contain a total of 10 carbons, and is therefore named as a bicyclodecane.
- Three bridges connect the bridgehead carbons: two contain four carbons, and one contains zero carbons. (The bond connecting the bridgehead carbons in a fused ring system is considered to be a bridge with zero carbons.)

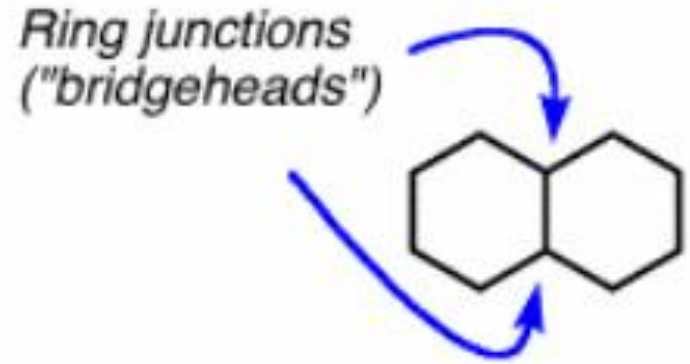


bicyclo[4.4.0]decane
(* = bridgehead carbons)

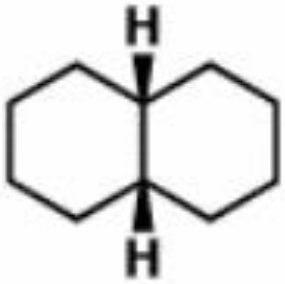
Fused Ring System
(*cis* and *trans* Ring Fusion)



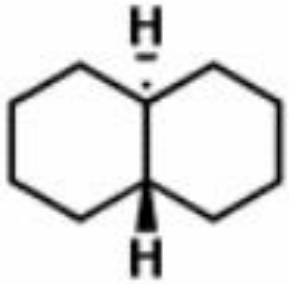
- Two rings in a fused bicyclic compound can be joined in more than one way.
- There are two stereoisomers of decalin: *cis*-decalin and *trans*-decalin



Decalin core structure
(ambiguous stereochemistry)

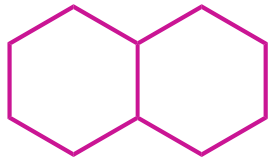


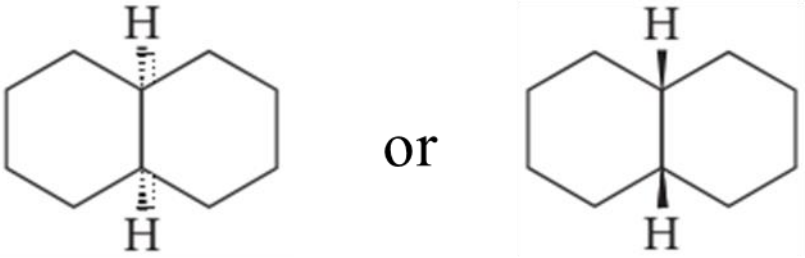
***cis*-decalin**



***trans*-decalin**

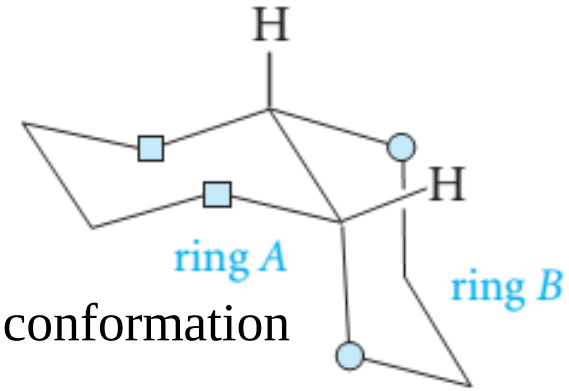
Fused Ring System
(*cis* and *trans* Ring Fusion)



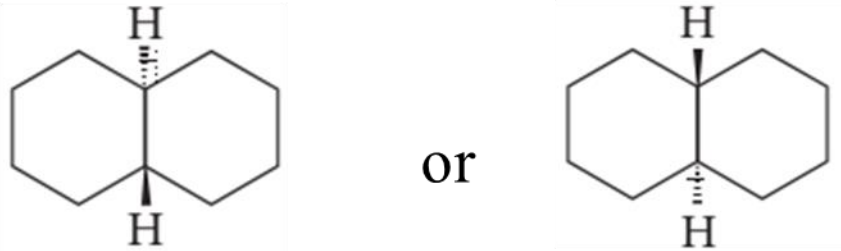


Planar structure

cis-decalin

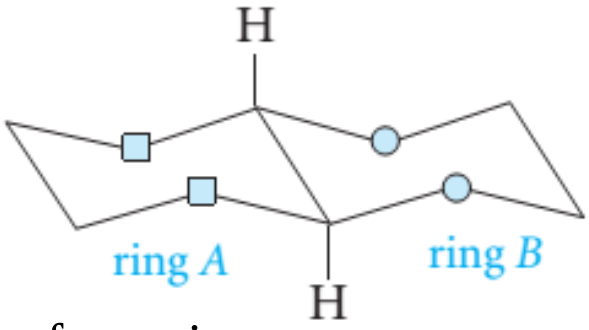


Chair conformation



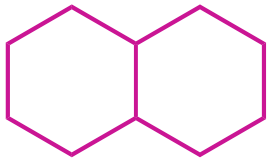
Planar structure

trans-decalin



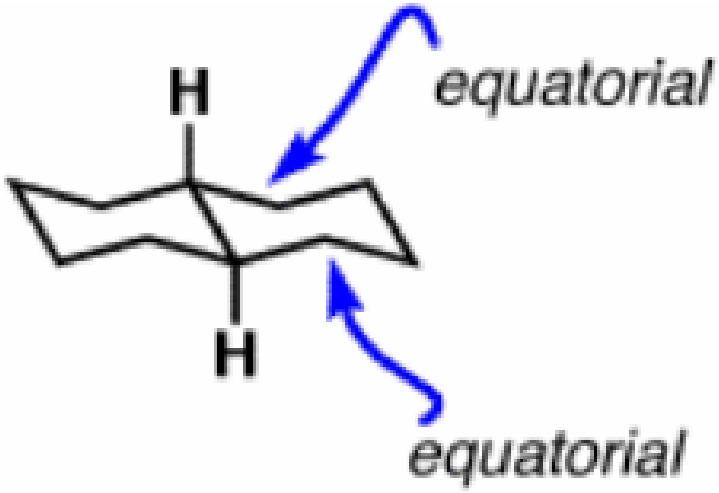
Chair conformation

Fused Ring System
(*cis* and *trans* Ring Fusion)

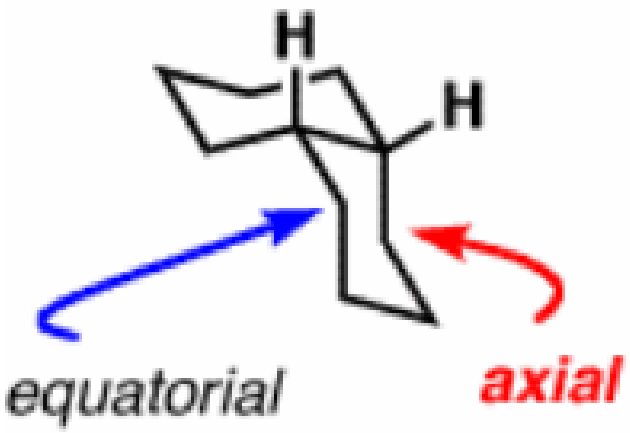


- Which one should be more stable, *cis*-decalin or *trans*-decalin?
- *trans*-decalin is more stable than *cis*-decalin since all carbons are equatorial whereas in *cis*-decalin, one carbon is equatorial and the other is axial.

trans-decalin



cis-decalin



Quiz

1. Draw the planar structure of decalin with the numbering of all carbon atoms and answer the following questions:

   i. Is the compound a fused or a bridged ring bicyclic compound?

   ii. Write another two names of the compound.

2. Decalin exists in two stereoisomeric forms:

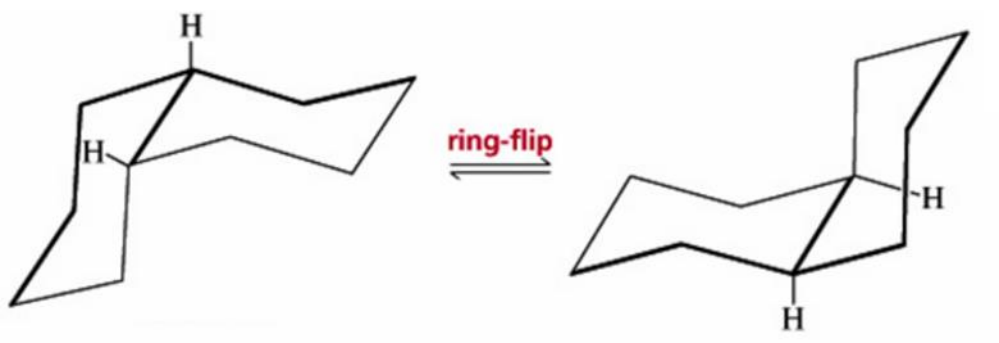
   i. What are the names of these two stereoisomeric forms of decalin?

   ii. Draw the planar structures and the corresponding chair conformations of the two stereoisomeric forms.

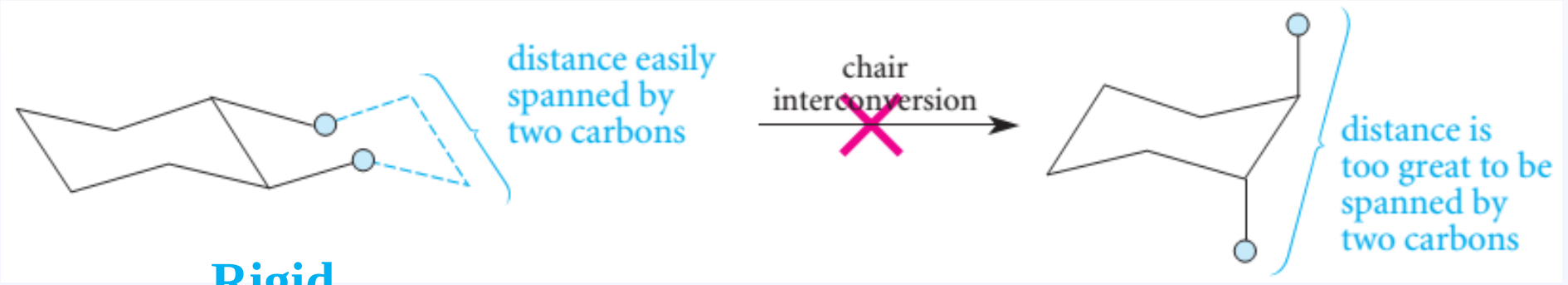
   iii. Which one of the stereoisomeric forms of decalin is more stable? Give explanation in support of your answer.

Fused Ring System
(chair interconversion)

- Each cyclohexane ring in *cis*-decalin can undergo the chair interconversion.



- However, in *trans*-decalin, the six-membered rings can assume twist-boat conformations, but they cannot change into their alternative chair conformations.



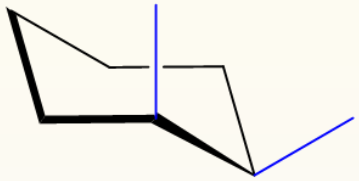
Rigid

cis- and *trans*-Decalin

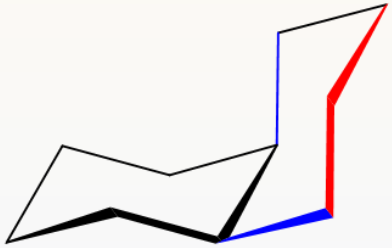


chair

interconversion



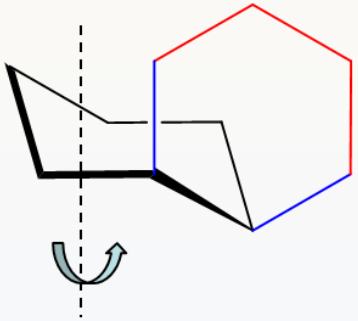
cis-1,2-Dimethylcyclohexane

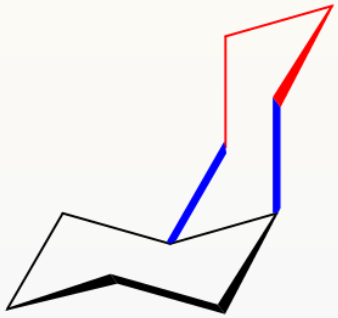


cis-Decalin

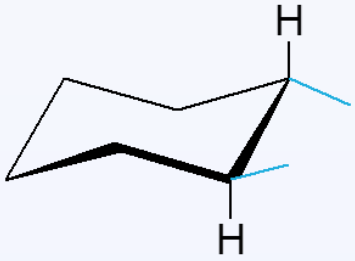
chair

interconversion

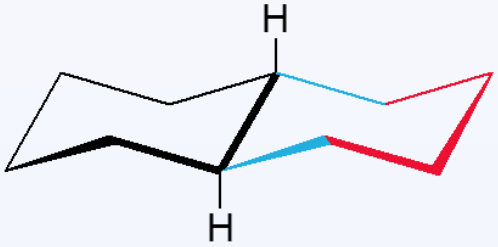




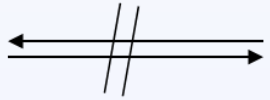
cis-Decalin



trans-1,2-Dimethylcyclohexane



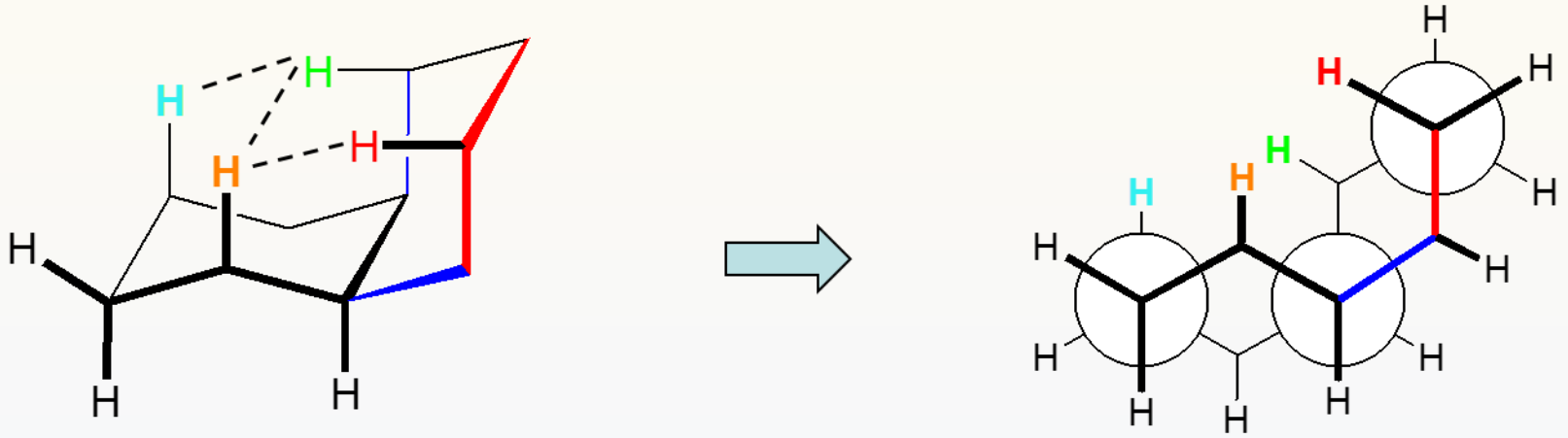
trans-Decalin



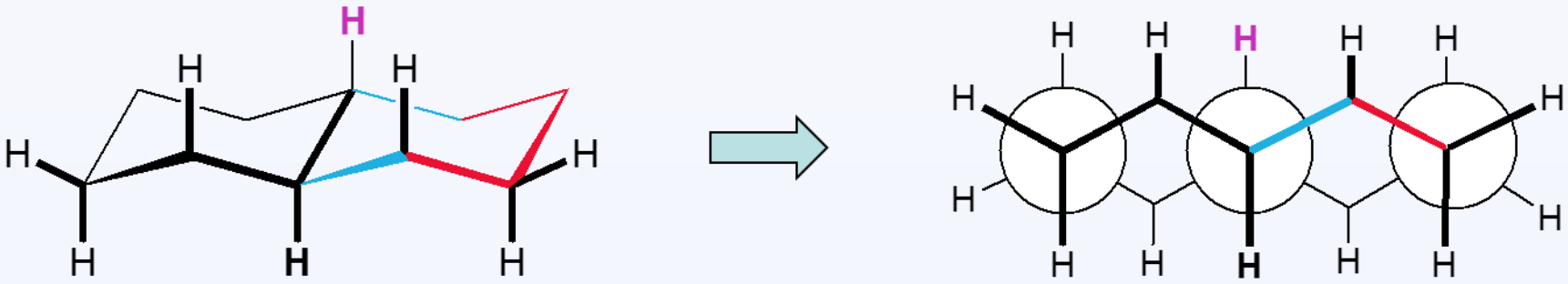
No chair interconversion

cis- and *trans*-Decalin

- *trans*-Decalin is more stable than *cis*-decalin. There are three extra-annular gauche interactions in *cis*-decalin. *cis*-Decalin, therefore, has a higher energy content than *trans*-decalin.



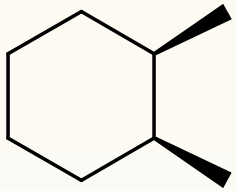
cis-Decalin



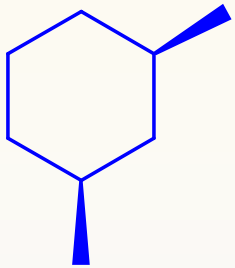
trans-Decalin

Chirality of Cyclic Compound

- Comment on the chirality of the following dimethylcyclohexanes:



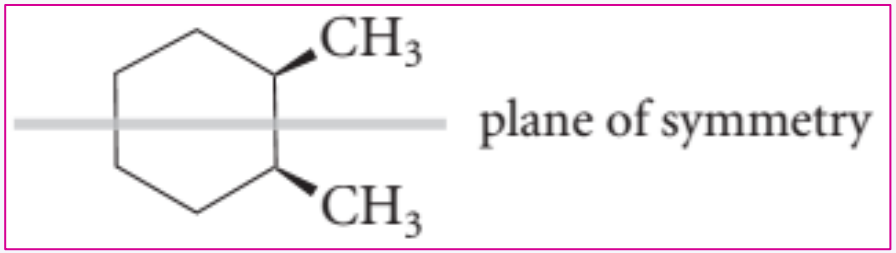
***cis*-1,2-isomer**



***cis*-1,3-isomer**



***cis*-1,4-isomer**

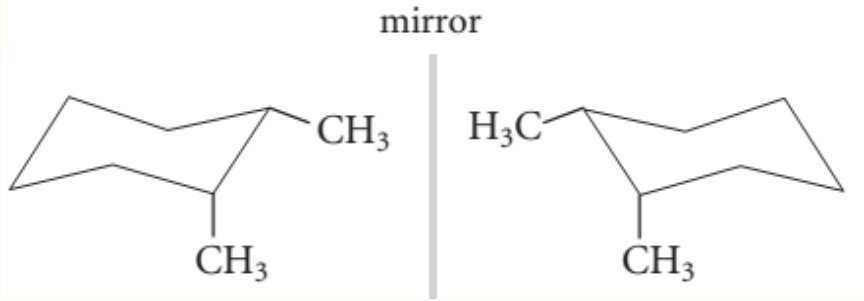


- The planar structures of *cis*-1,2-dimethylcyclohexane have internal mirror planes (planes of symmetry).

- However, this is not a plane of symmetry in the case of *cis*-1,2-dimethylcyclohexane, because each conformation of *cis*-1,2-dimethylcyclohexane is chiral and has no symmetry plane.

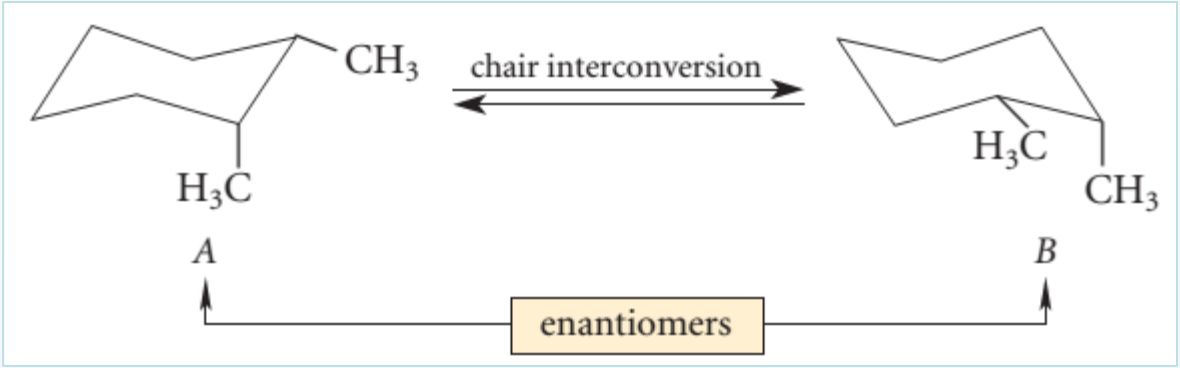
Stereochemical Consequences of the Chair Interconversion

- *cis*-1,2-dimethylcyclohexane is chiral

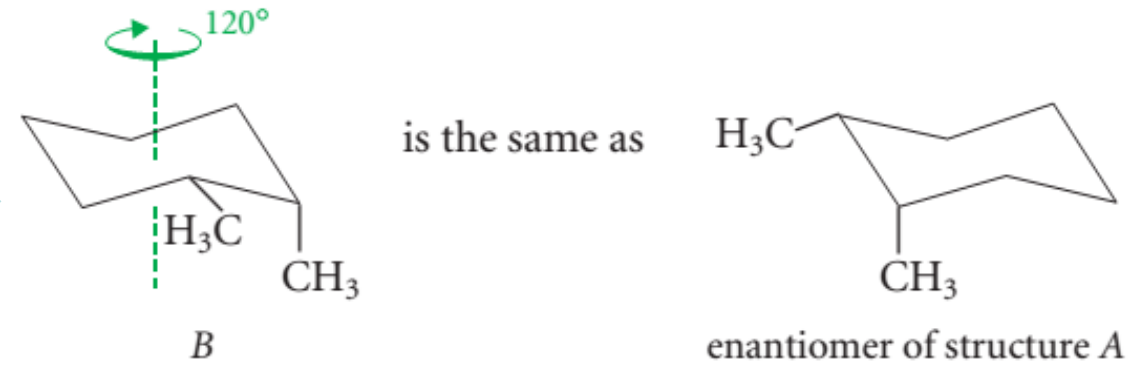


Non-congruent mirror images of each other

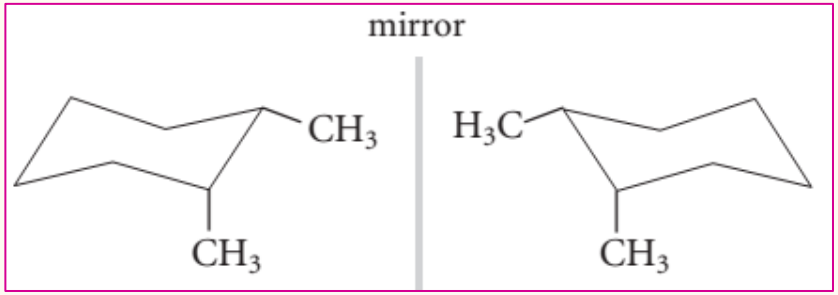
- However, the chair interconversion converts one enantiomer into the other:

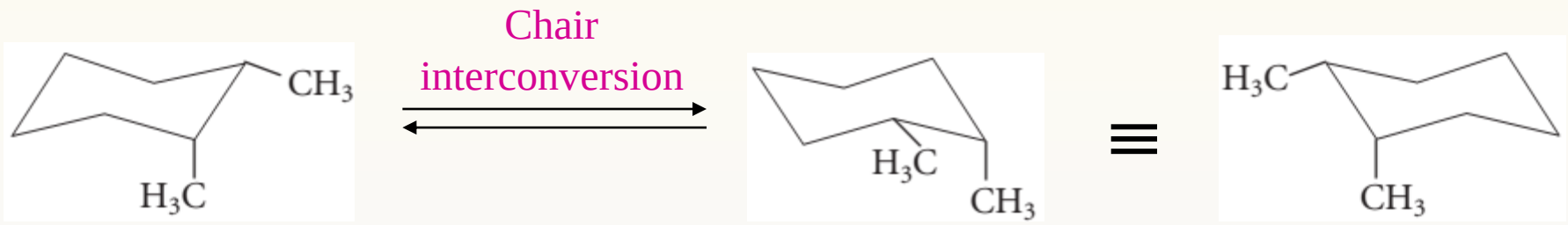


- By turning structure B 120° about a vertical axis :



Chirality of *cis*-1,2-Dimethylcyclohexane

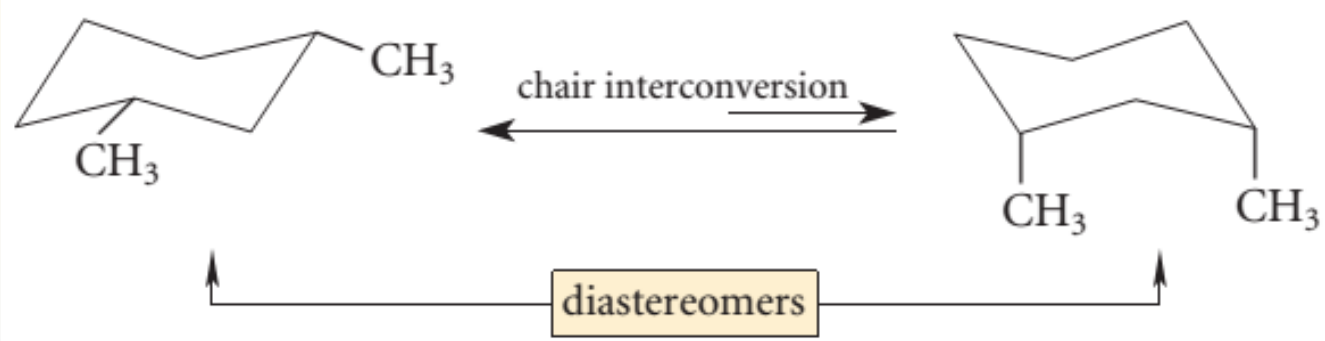




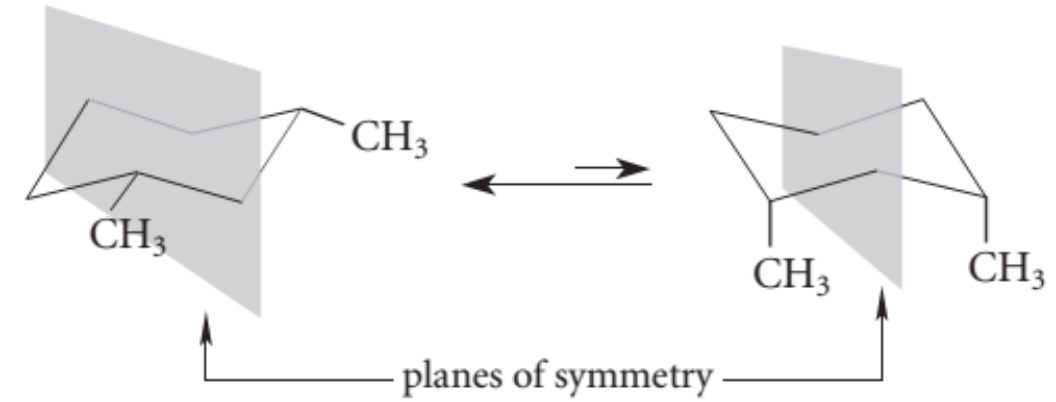
- *cis*-1,2-dimethylcyclohexane is a mixture of chiral (and enantiomeric) conformations in rapid equilibrium.
- Because these enantiomers are interconverted very rapidly, they cannot be separated at ordinary temperatures.
- Although it does not exhibit optical activity at ordinary temperatures, at a temperature so low that the rate of the chair interconversion would be negligible, it could in principle be separated into enantiomers, each of which would be optically active.

Chirality of *cis*-1,3-Dimethylcyclohexane

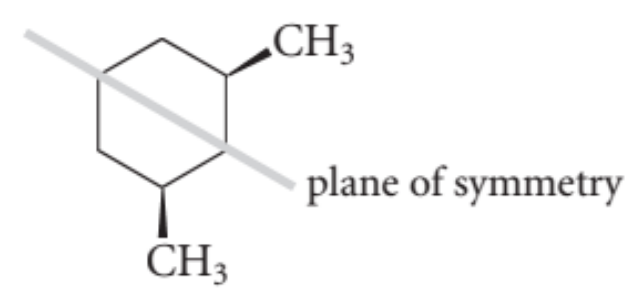
- *cis*-1,3-dimethylcyclohexane, another molecule with two asymmetric carbons, is a mixture of conformational diastereomers:

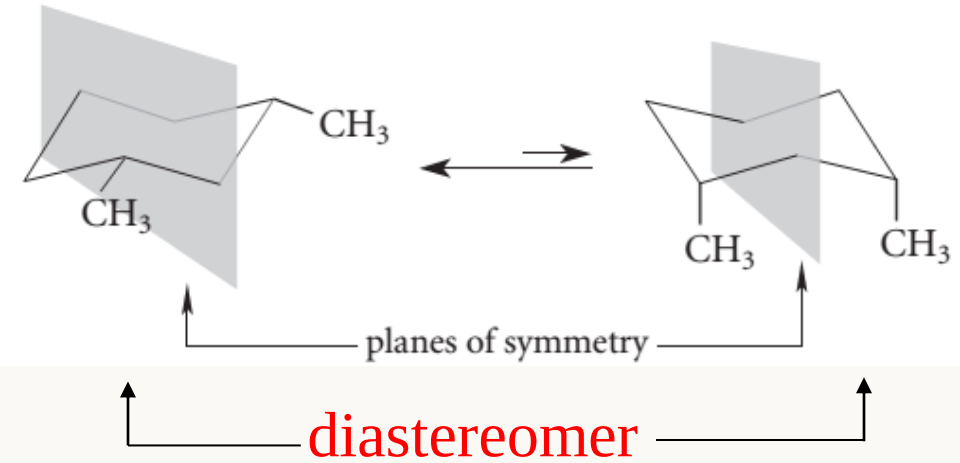


- Yet, even though each conformation has two asymmetric carbons, neither conformation is chiral because each has an internal plane of symmetry.



Chirality of cis-1,3-Dimethylcyclohexane

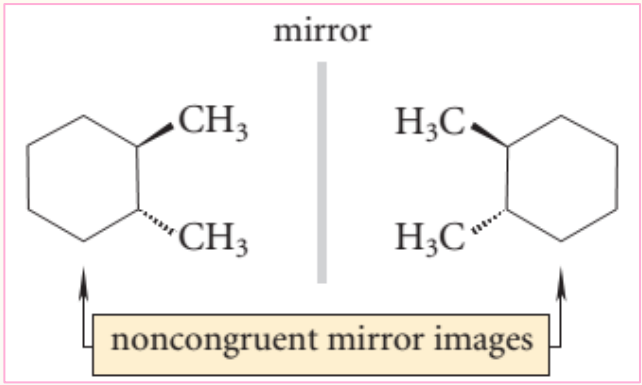




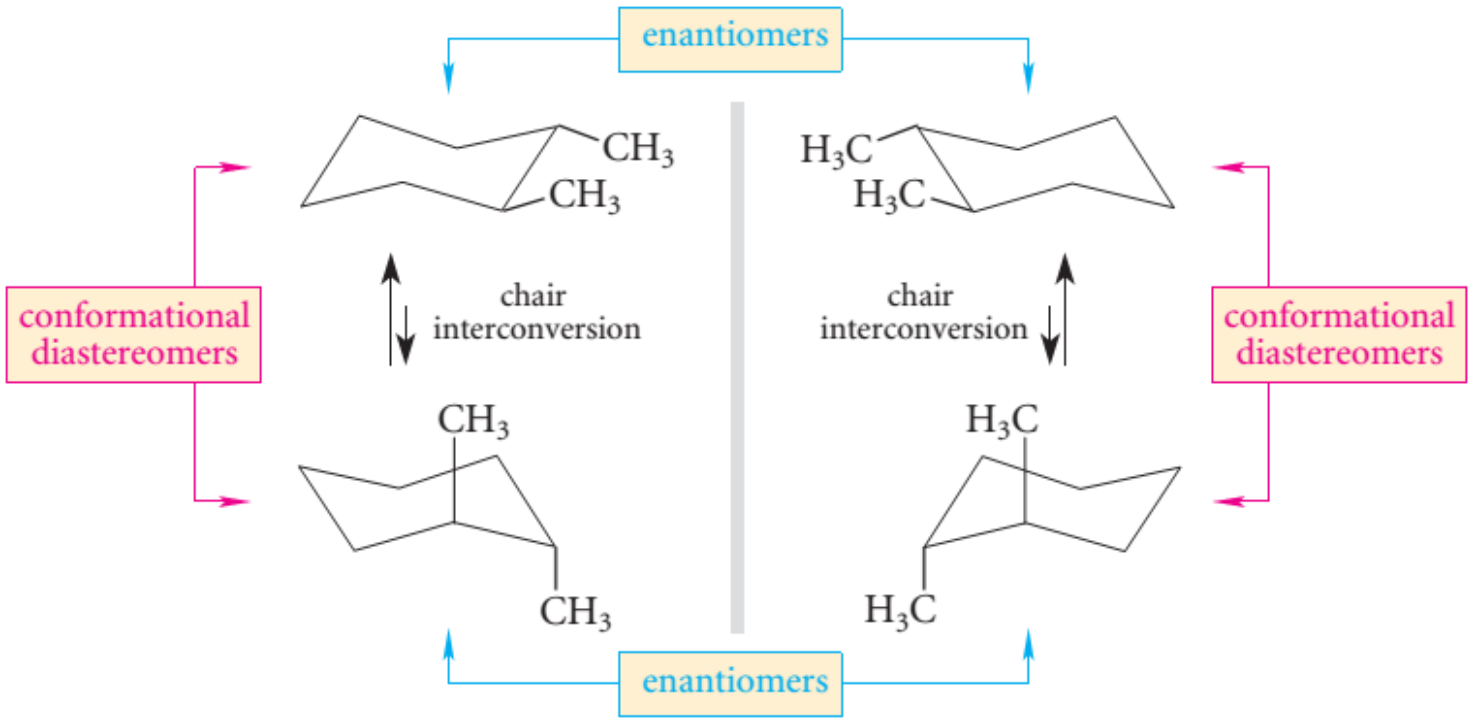
- Both conformations of *cis*-1,3-dimethylcyclohexane are meso.
- *cis*-1,3-dimethylcyclohexane is thus a rapidly interconverting mixture of two different meso compounds, each a diastereomer of the other.
- Like any meso compound, *cis*-1,3-dimethylcyclohexane is achiral *i.e.* optically inactive.

Chirality of *trans*-1,2-Dimethylcyclohexane

- We can see from its planar structures that *trans*-1,2-dimethylcyclohexane is chiral.
- Each enantiomer is capable of independent existence and can be isolated in optically active form.

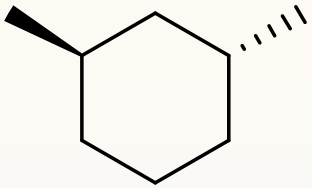


- The chair interconversion converts each enantiomer into a conformational diastereomer.

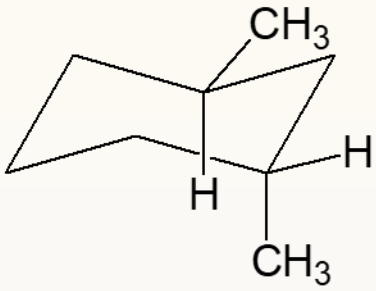


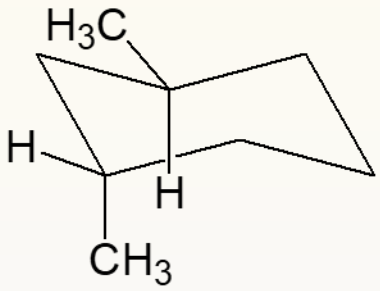
Chirality of *trans*-1,3-Dimethylcyclohexane

- *trans*-1,3-dimethylcyclohexane does not have a plane of symmetry and exists as a pair of enantiomers.

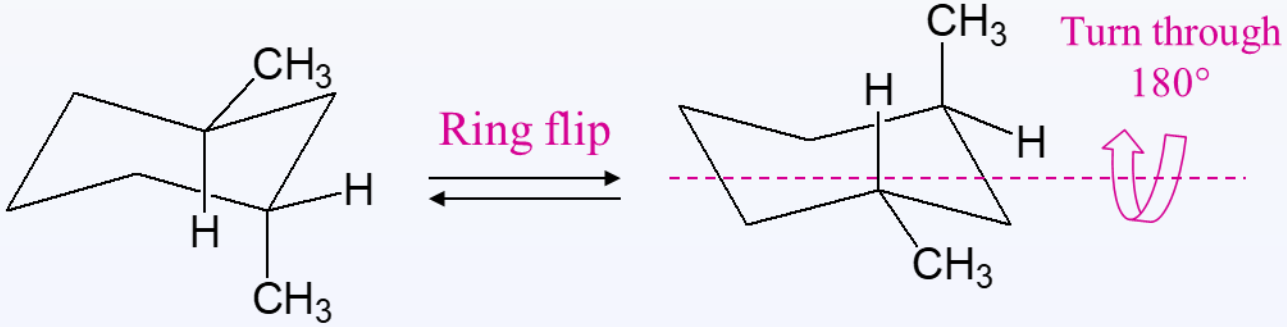


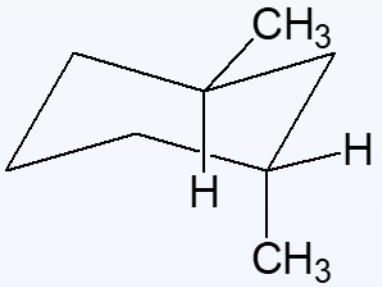
Chiral





- The ring flip in *trans*-1,3-dimethylcyclohexane does not racemize this compound but regenerate the same enantiomer.

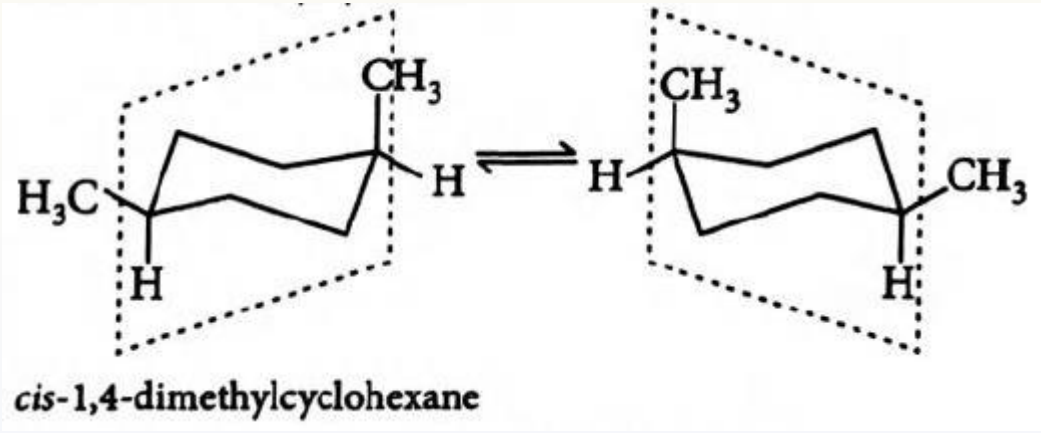




- The ring flip regenerate the same enantiomer.

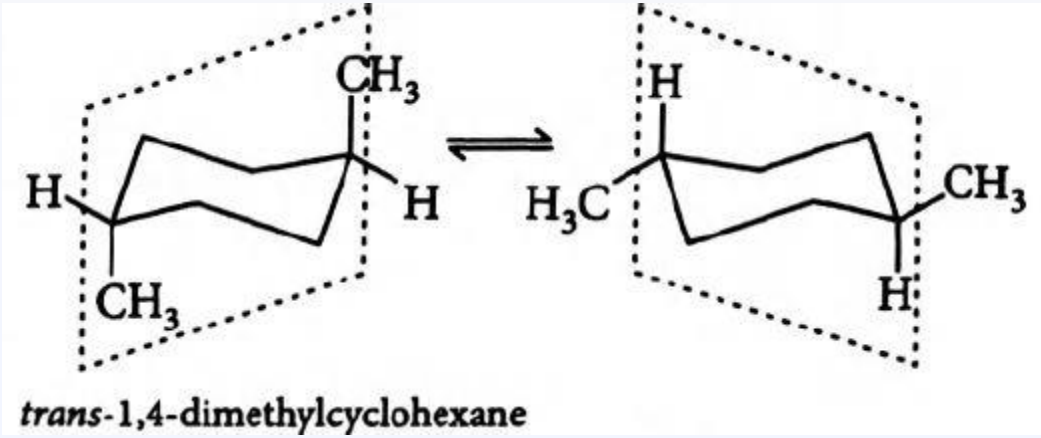
Chirality of *cis*- and *trans*-1,4-Dimethylcyclohexane

- *cis*- and *trans* - 1,4-dimethylcyclohexane contain an internal symmetry plane, only two stereoisomers appear. Therefore, they are both meso compounds (optically inactive)



cis-1,4-dimethylcyclohexane

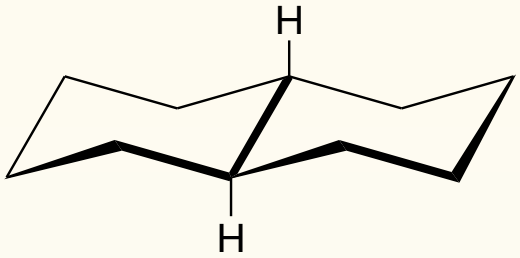
meso compounds
(optically inactive)



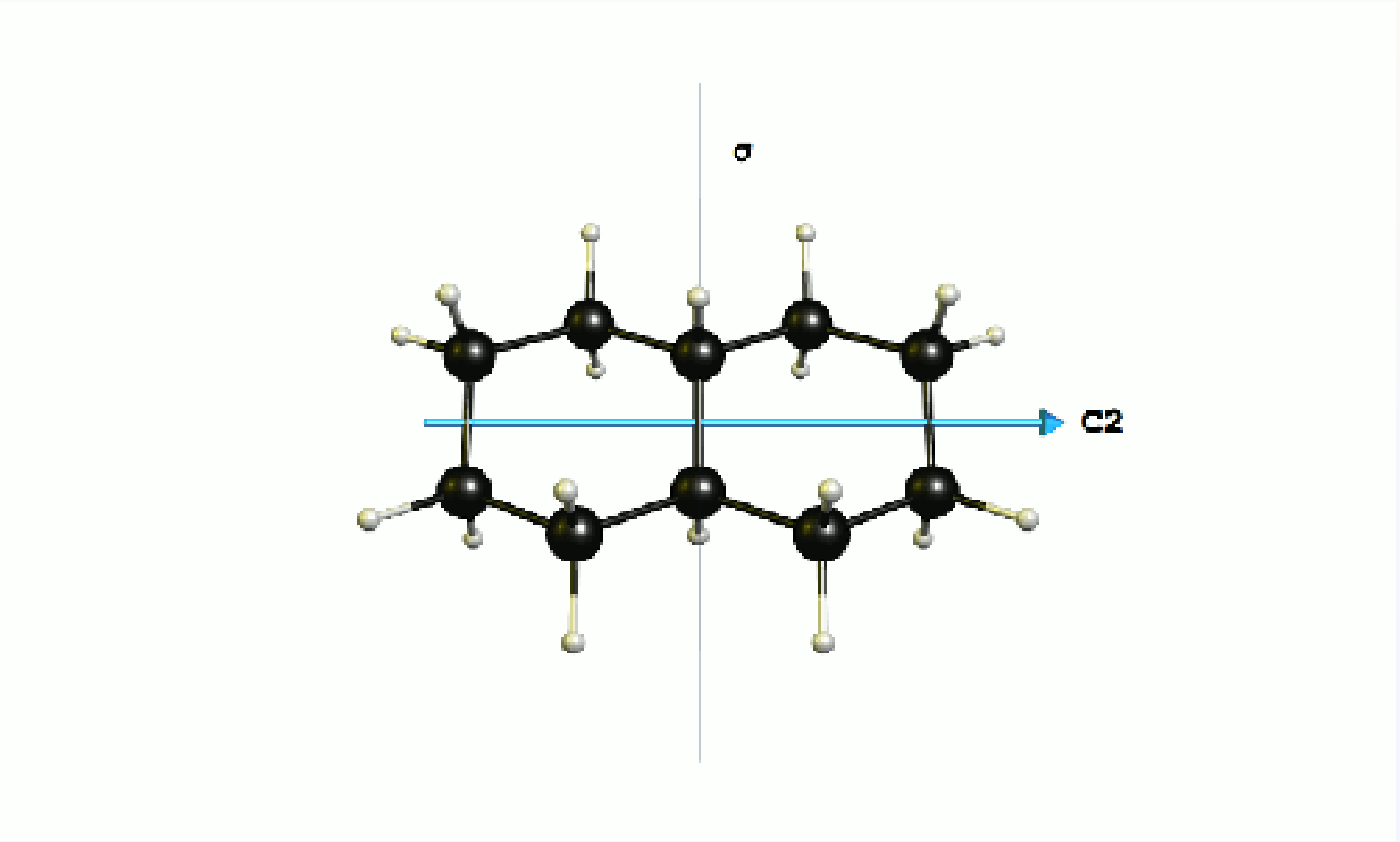
trans-1,4-dimethylcyclohexane

meso compounds
(optically inactive)

Chirality of *trans*-Decalin

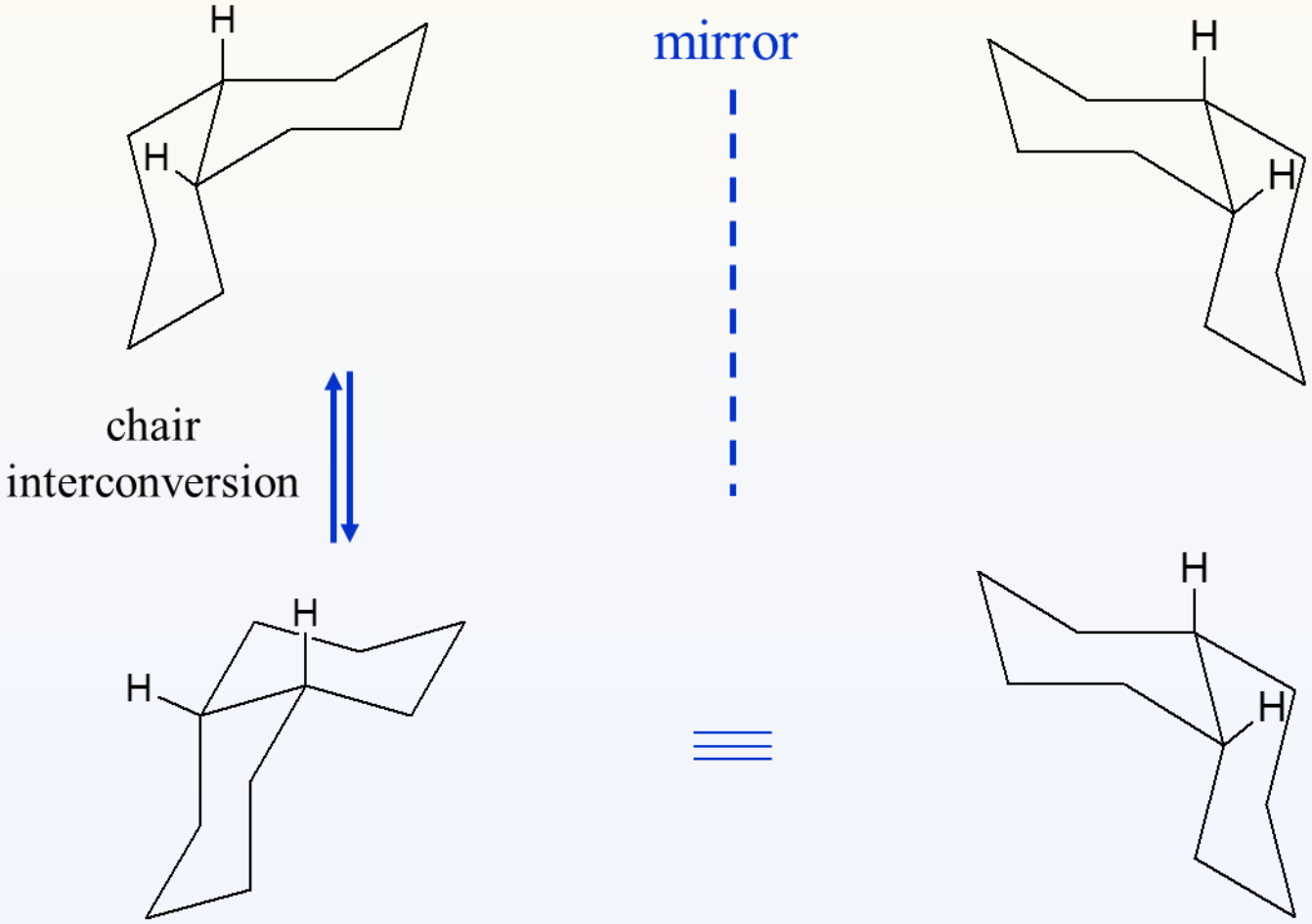


- The symmetry point group of *trans*-decalin is C_{2h}.
- *trans*-Decalin is not optically active, it is a meso compound.



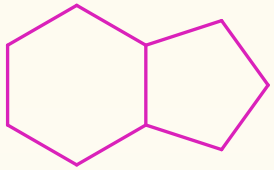
Chirality of *cis*-Decalin

- *cis*-Decalin is chiral and exists as a pair of non-resolvable enantiomers.
- Since the chair interconversion converts one enantiomer into the other, they cannot be separated at ordinary temperatures.



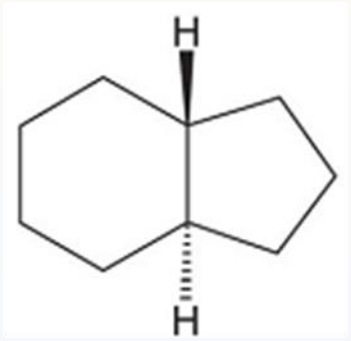
Stereochemistry Of Fused And Bridged Ring System

Fused Ring System (Hydrindane)

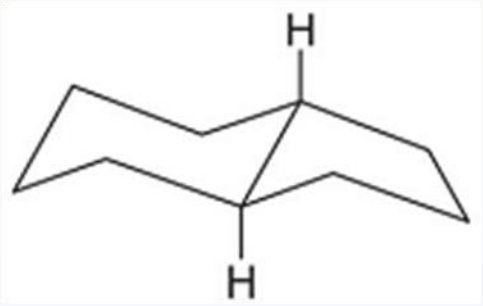


bicyclo[4.3.0]nonane

- Hydrindane (octahydro-1*H*-indene) is an interesting molecule because it results from the fusion of two rings of different size, and also because it represents a portion (rings C and D) of the steroid nucleus.

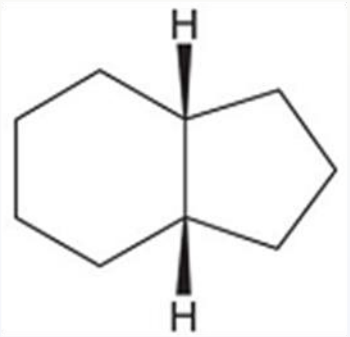


Planar form

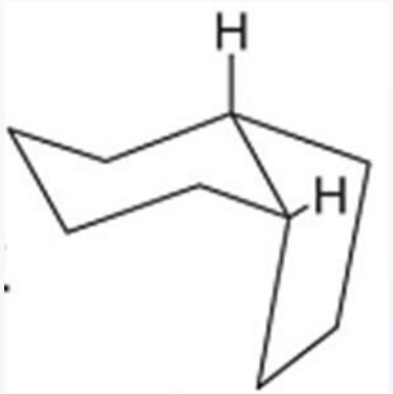


Chair form

trans-Hydrindane



Planar form

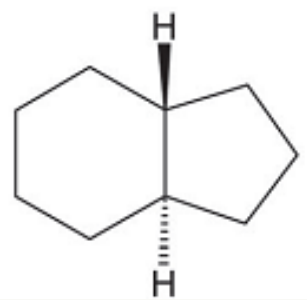


Chair form

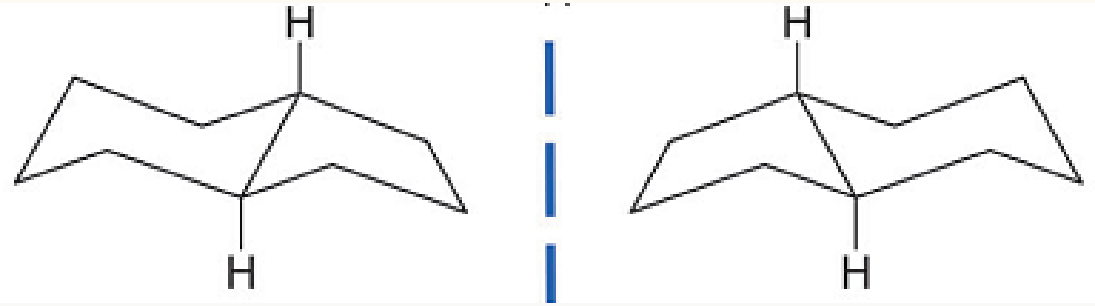
cis-Hydrindane

Chirality of *cis*- and *trans*-Hydrindanes

- The *trans*-hydrindanes are conformationally rigid molecules and occurs as two stable *trans*-enantiomers.

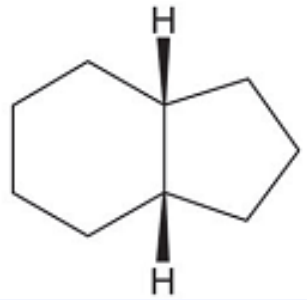


trans-Hydrindane

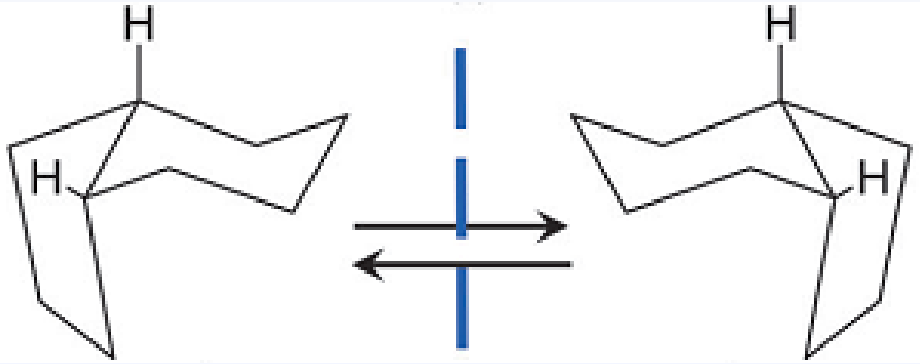


trans-(*R*,*R*)- *trans*-(*S*,*S*)-

- **The *cis*-hydrindanes is flexible and occurs as a pair of enantiomeric conformers.**

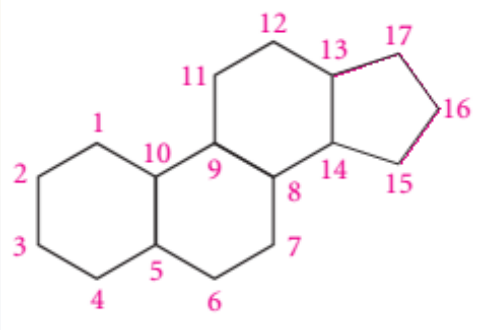


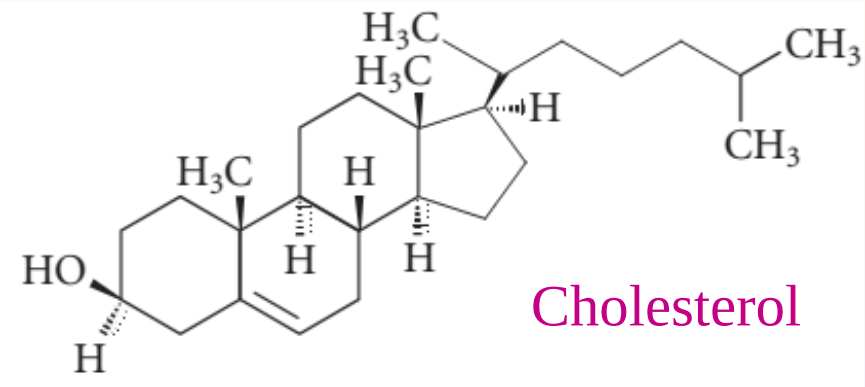
cis-Hydrindane (meso)



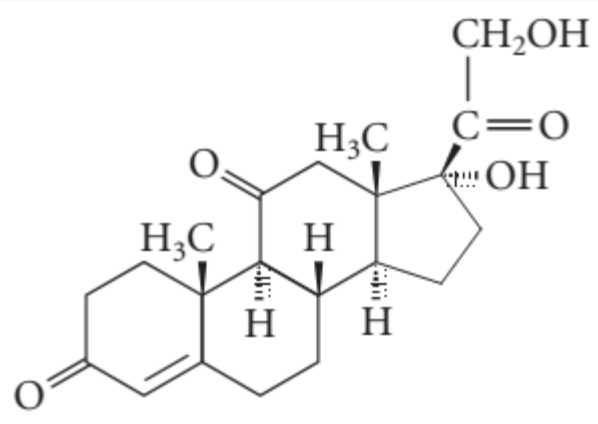
Steroids

- Of the many naturally occurring compounds with fused rings, the steroids are particularly important.
- A **steroid** is a compound with a structure derived from the following tetracyclic ring system.

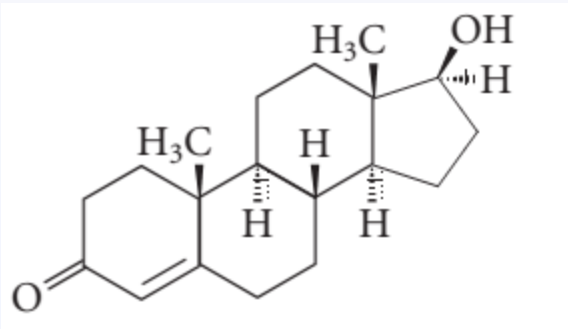




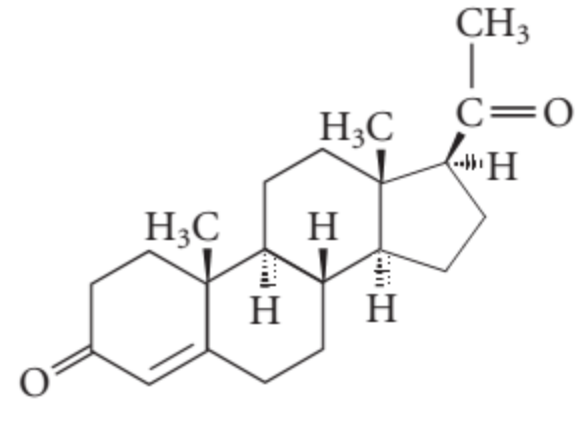
Cholesterol



cortisone
(anti-inflammatory hormone)



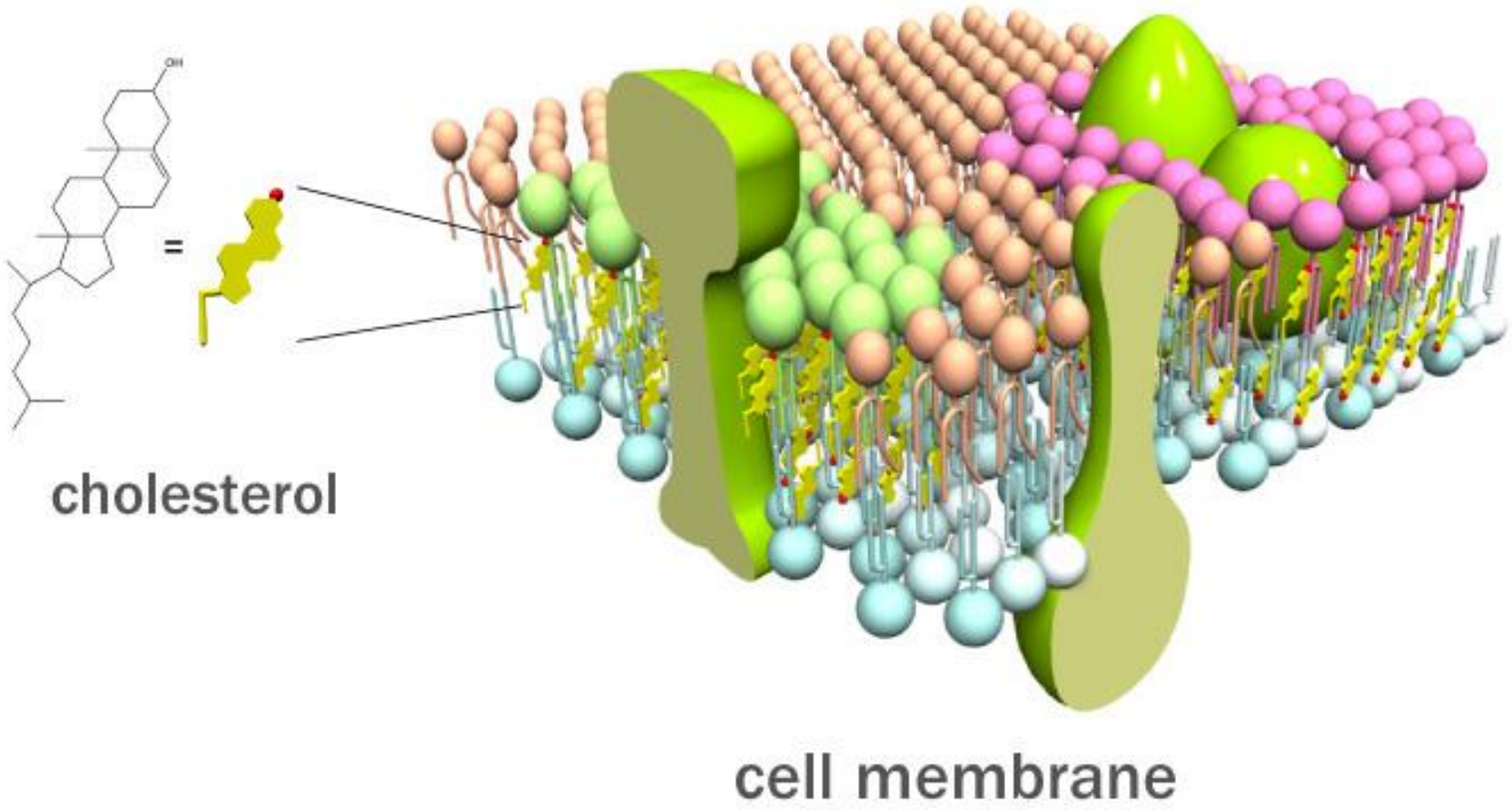
testosterone
(human male sex hormone)



progesterone
(human female sex hormone)

Cholesterol

- Important cell component



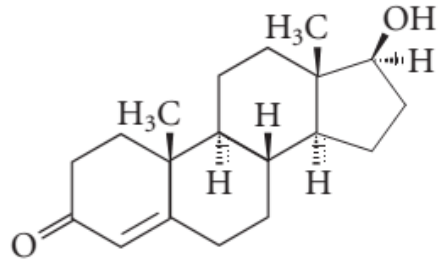
Cholesterol

- Precursor of all other steroids including vertebrate sex hormones.

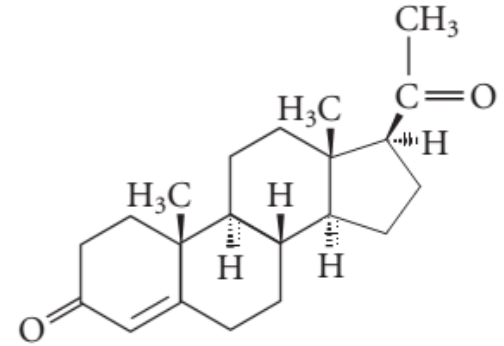
Cholesterol → Sex Hormones







testosterone
(human male sex hormone)



progesterone
(human female sex hormone)

What a big difference a few atoms can make!

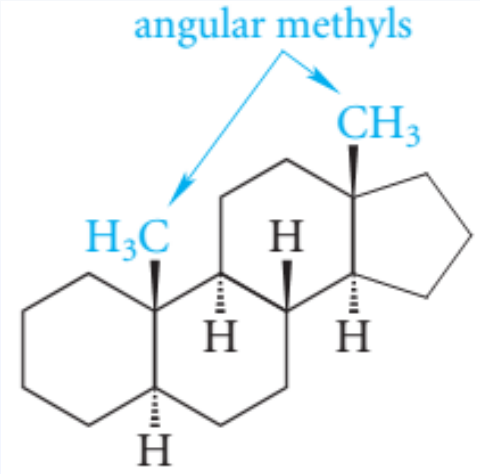
Steroids

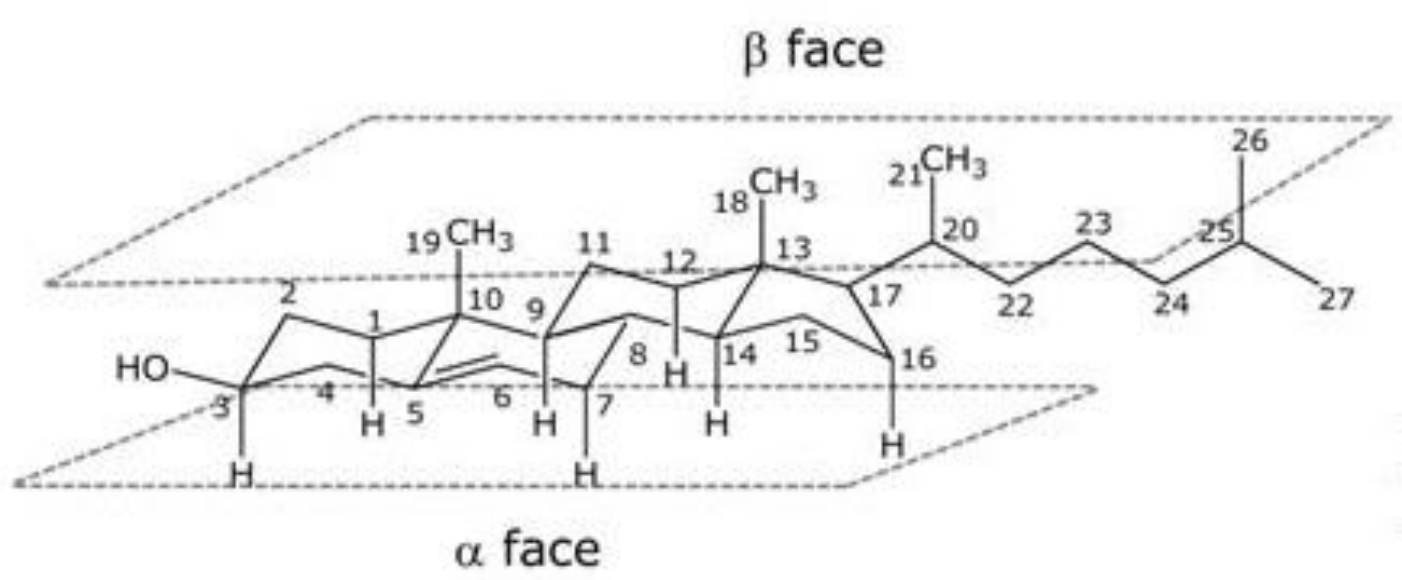
- Two structural features are particularly common in naturally occurring steroids:

  i. The first is that in many cases all ring fusions are trans.

  ➢ Because trans-fused cyclohexane rings cannot undergo the chair interconversion, all-trans ring fusion causes a steroid to be conformationally rigid and relatively flat.

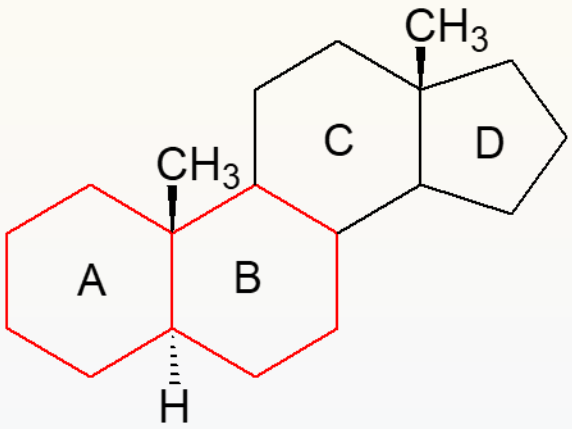
  ii. Many steroids have methyl groups, called angular methyls, at carbons 10 and 13.





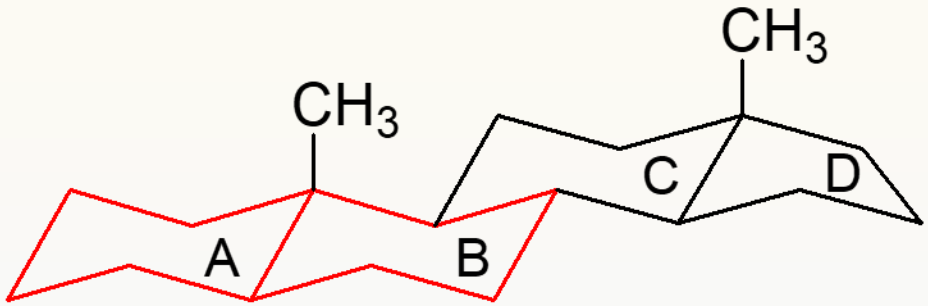
Common Steroid Conformations

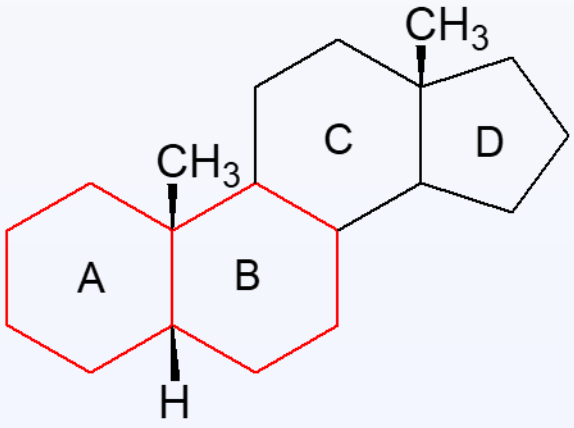
A/B ring fusion may be *cis* or *trans*



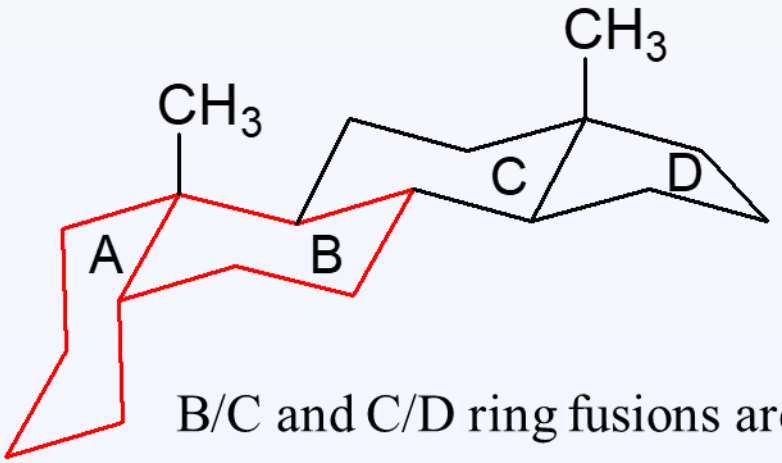
A/B ring fusion is *trans*

B/C and C/D ring fusions are *trans*





A/B ring fusion is *cis*



B/C and C/D ring fusions are *trans*

Alkaloids

Strychnine - A Highly Toxic Indole Alkaloid





- Strychnine is most commonly derived from the seeds of *Strychnos nux-vomica*, a native tree of South India.



- Strychnine injections are used to kill stray dogs/ seeds to kill animals, now found in pesticides (kill rats).
- **30 mg of strychnine is enough to be fatal to an adult**.

Alkaloids

Conformation of Strychnine:

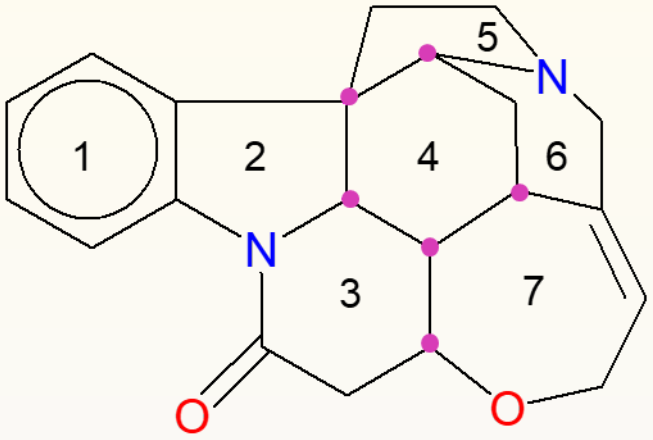
i. Ring 4 is in chair conformation, thus forcing the bridging of ring 6 with it via *cis*-diaxial bonds.

ii. This makes the fusion of ring 5 with 6 via one axial and one equatorial bond i.e. *cis*-fusion.

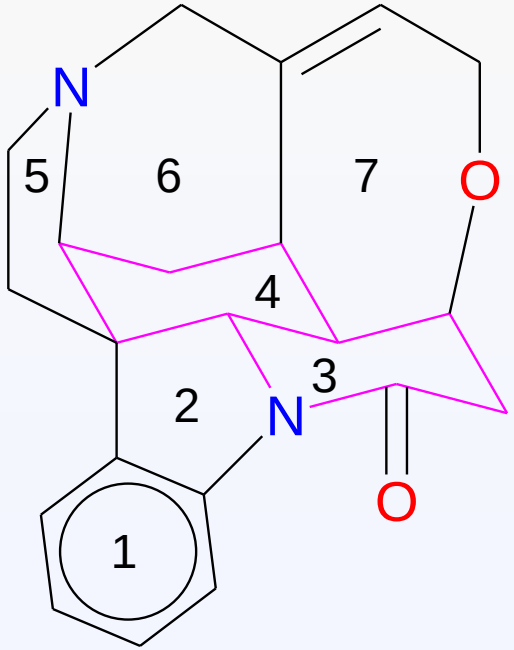
iii. Ring 2 is also *cis*-fused with ring 4 *i.e.*, its bond with nitrogen is equatorial.

iv. Ring 7 is also *cis*-fused, and

v. The fusion of ring 3 and 4 is *trans*.

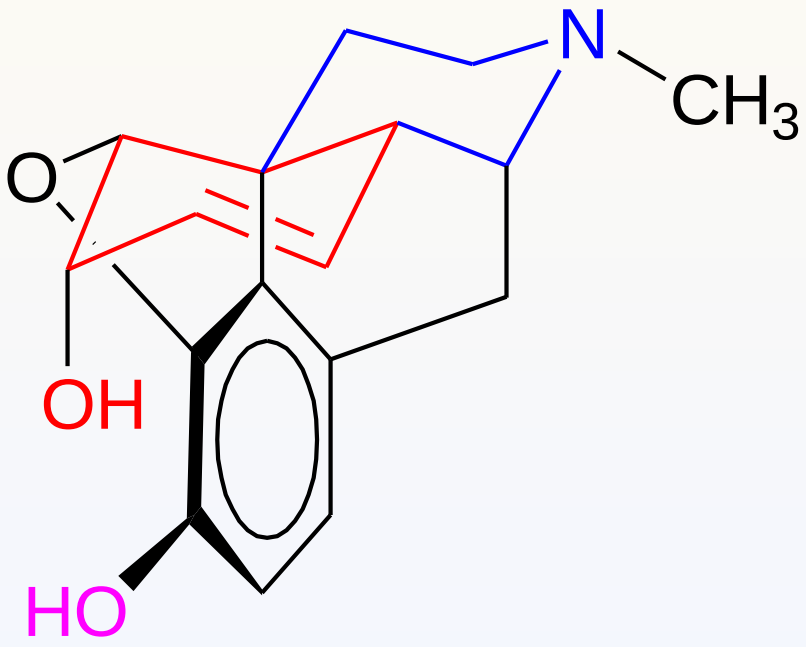


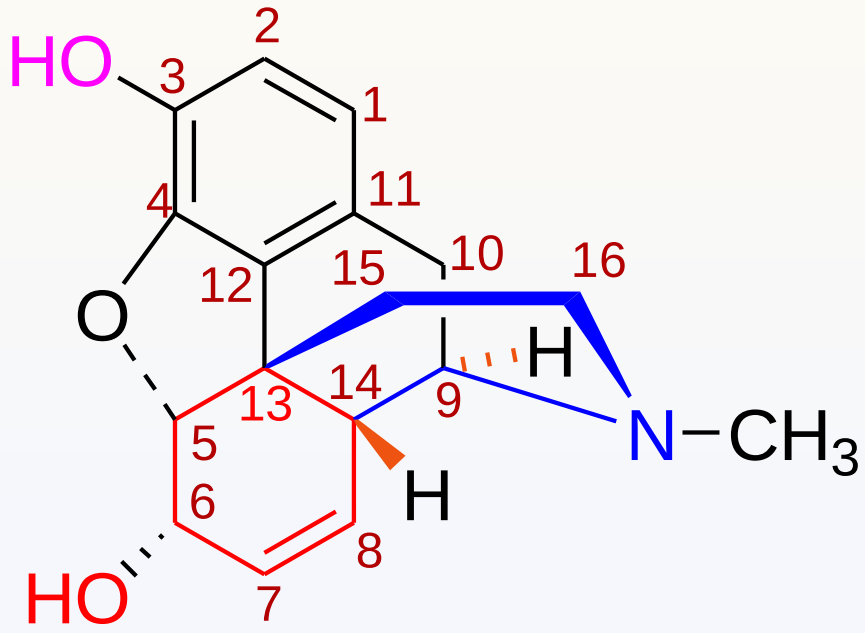
Strychnine



Alkaloids

- The conformation formula of morphine is drawn as follows:

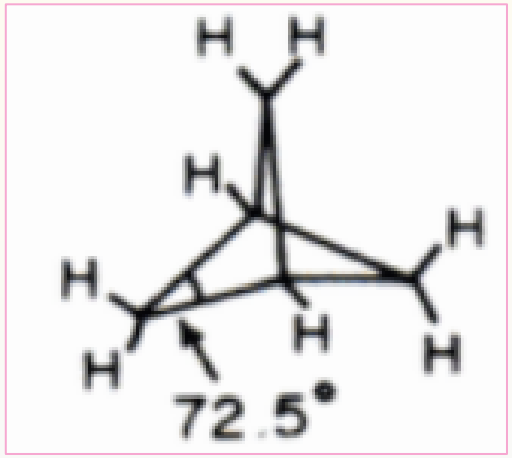


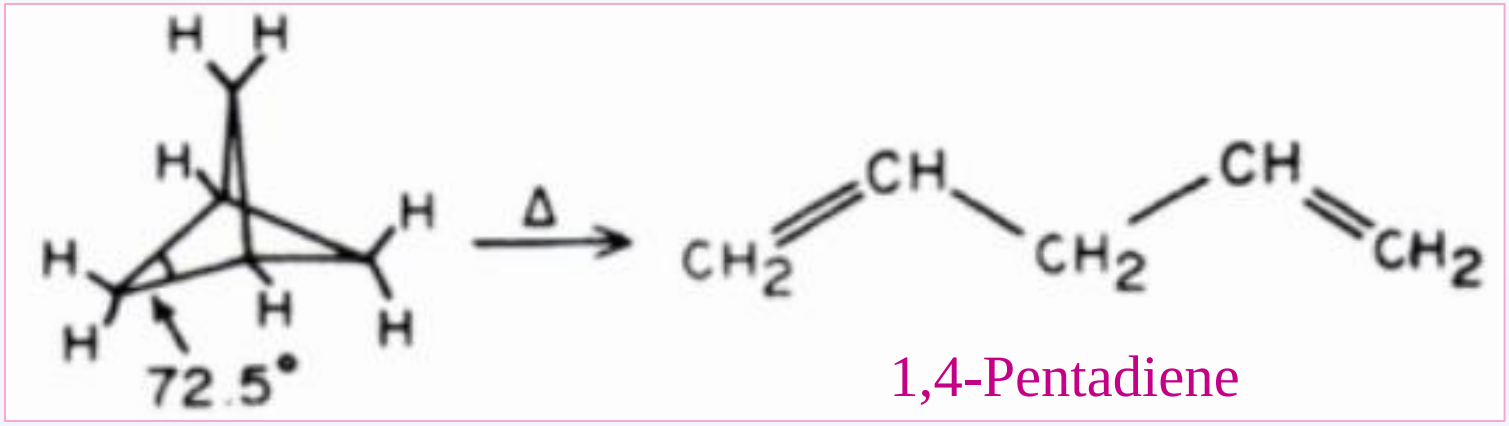


Morphine

Bridged Ring Systems

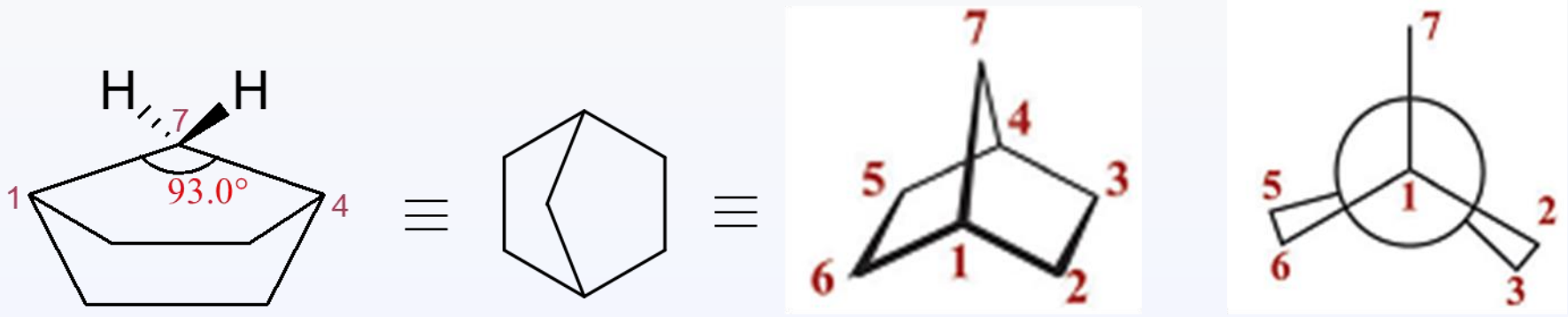
- The smallest bridged ring system is bicyclo[1.1.1]pentane.
- Two bridgehead carbons are brought to close proximity because of very small endocyclic valency angles (72.5°).
- It is unstable and converted into 1,4-pentadiene on heating at 300 °C.





Bridged Ring Systems

- The most common bridged ring system is bicyclo[2.2.1]heptane or norbornane.
- In this system, C-1 and C-4 of cyclohexane are joined by a methylene bridge forcing it to adopt a boat conformation.
- The smallest endocyclic valency angle is 93.0°.
- The system is considerably strained because of two eclipsed butane units in the boat.

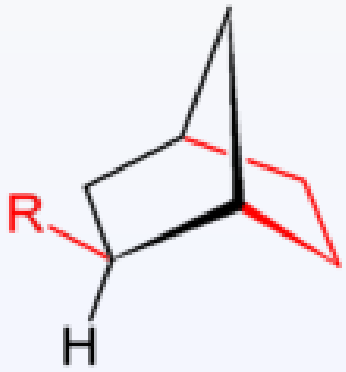


Bicyclo[2.2.1]heptane or norbornane

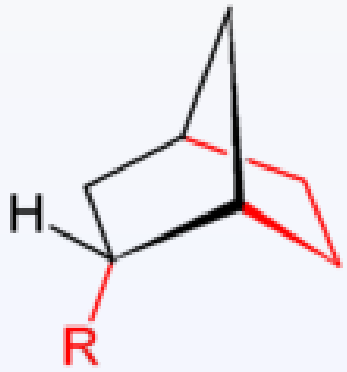
Bridged Ring Systems

Exo-Endo Isomerism

- ***Exo–endo*** **isomerism** is a special type of stereoisomerism found in organic compounds with a substituent on a bridged ring system.
- The prefix ***endo*** is reserved for the isomer with the substituent located closest, or "*syn*", to the longest bridge.
- The prefix ***exo*** is reserved for the isomer with the substituent located farthest, or "*anti*", to the longest bridge.



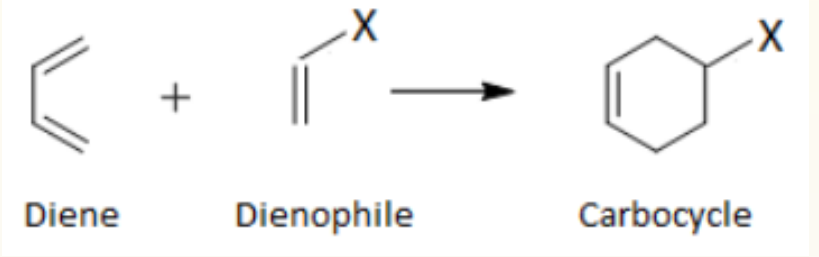
Exo

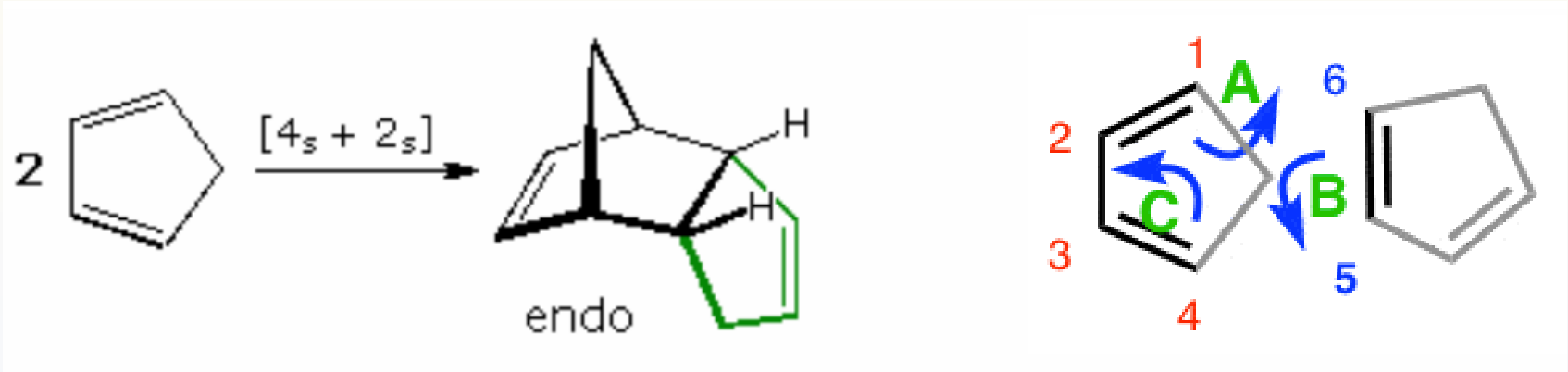


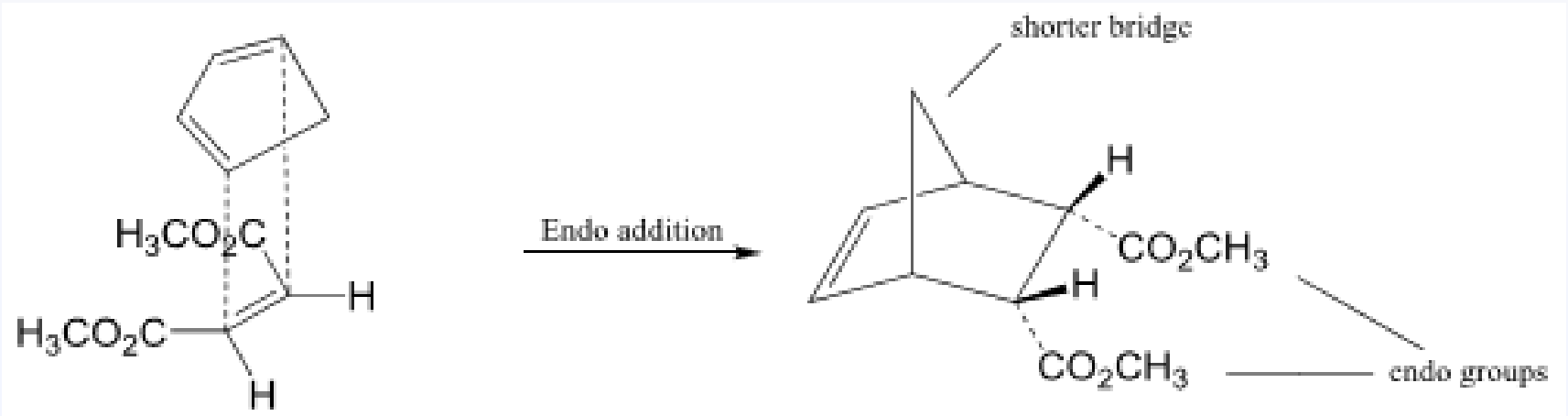
Endo

Bridged Ring Systems

- The Diels-Alder reaction usually proceeds with *endo selectivity*.



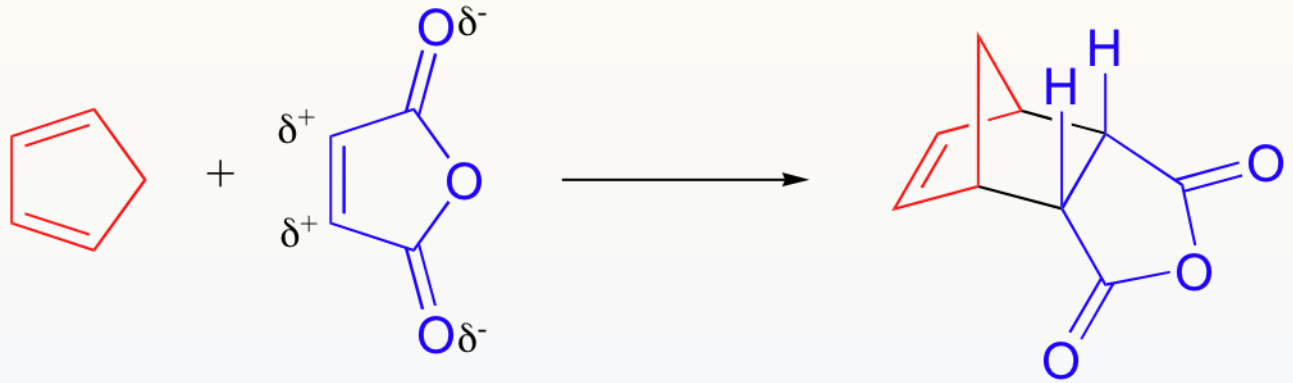




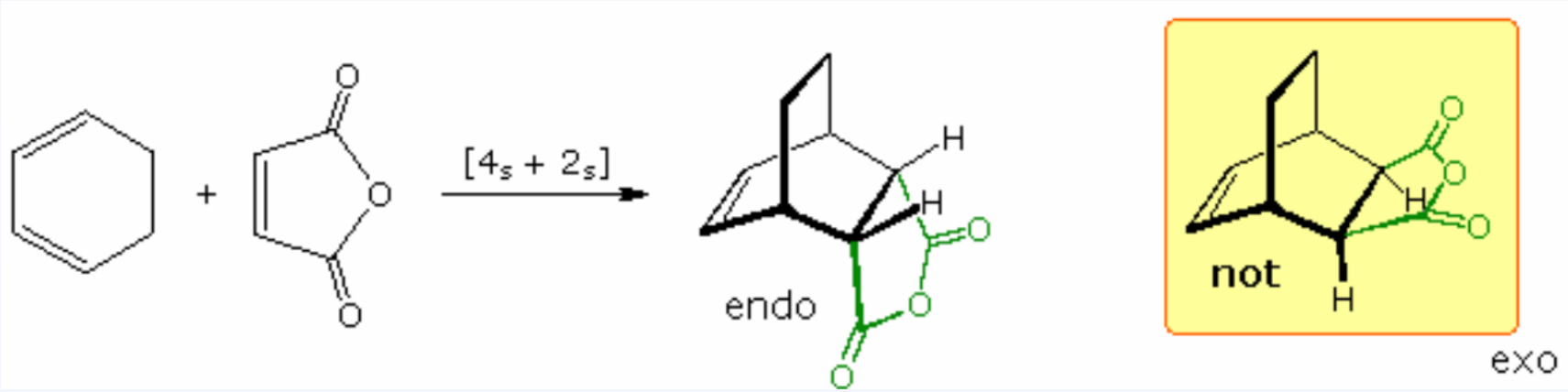
Bridged Ring Systems

Exo-Endo Isomerism

- In the Diels-Alder reaction, the stereochemistry of addition, besides being *endo, is also* cis.

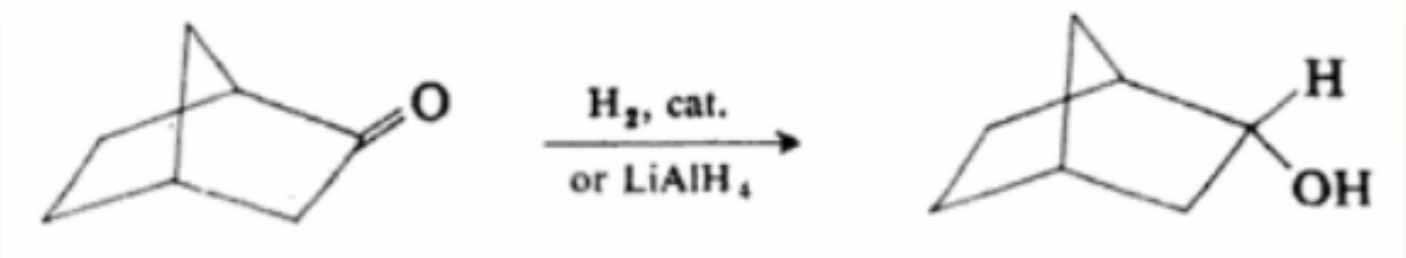


Cyclopentadiene Maleic anhydride *cis*-Norbornene-5,6-*endo*-Dicarboxylic Anhydride



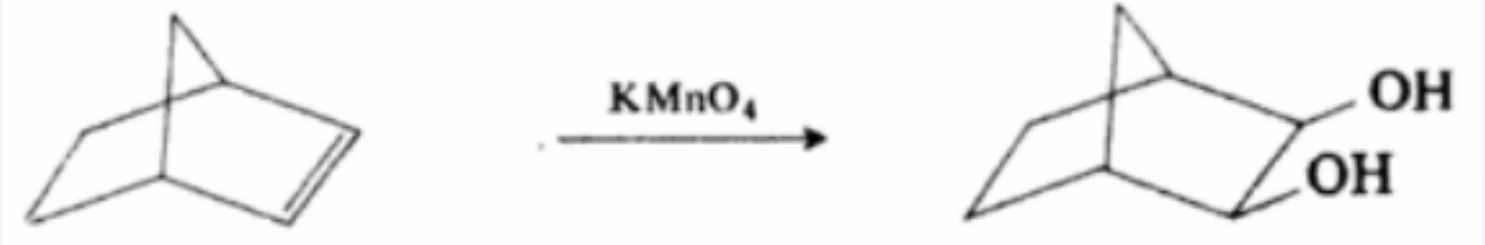
Bridged Ring Systems

- The endo side of the norbornane system is more sterically crowded than exo side ⇒ so the exo side is more accessible to a variety of reagents than the endo side.



Norcamphor

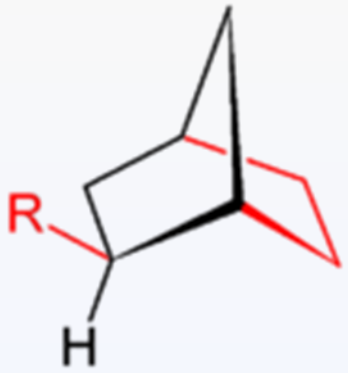
endo-Norborneol



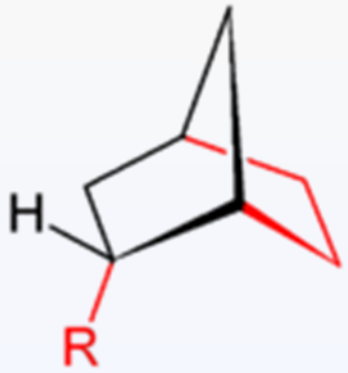
Norbornylene

exo-Norbornadiol

Stereochemistry Of Fused And Bridged Ring System



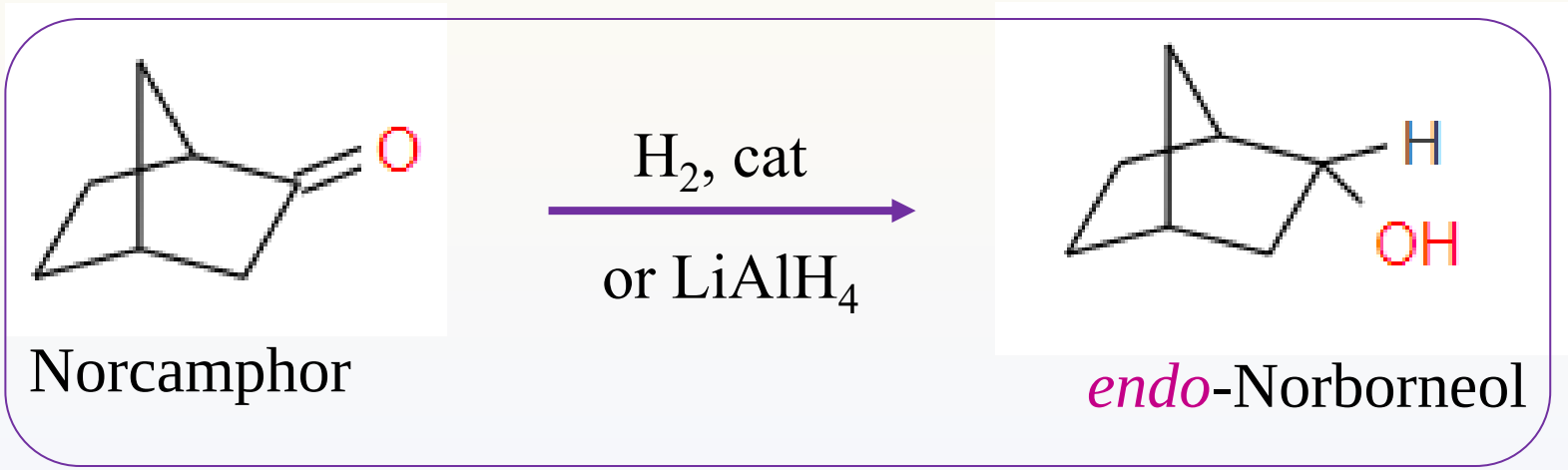
Exo



Endo

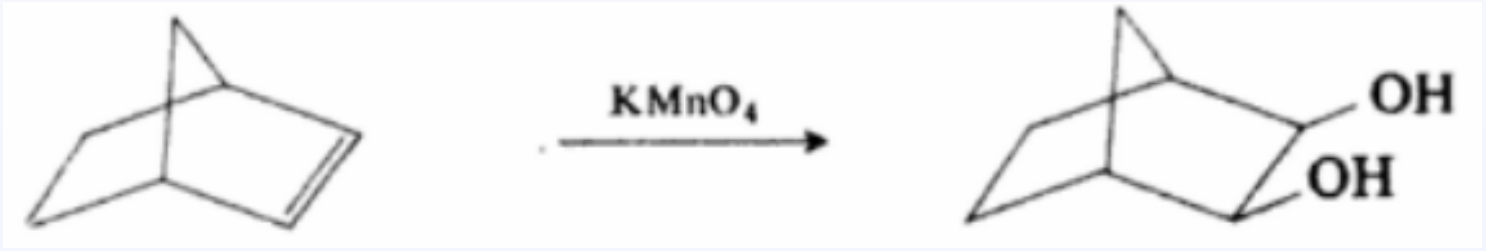
Bridged Ring Systems

- The endo side of the norbornane system is more sterically crowded than exo side ⇒ so the exo side is more accessible to a variety of reagents than the endo side.



Norcamphor

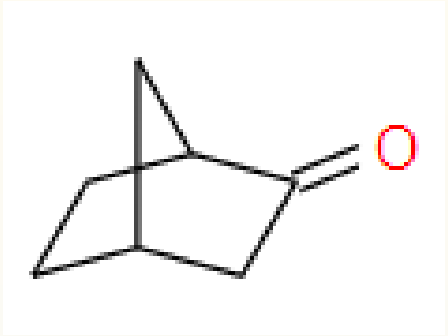
endo-Norborneol



Norbornylene

exo-Norbornadiol

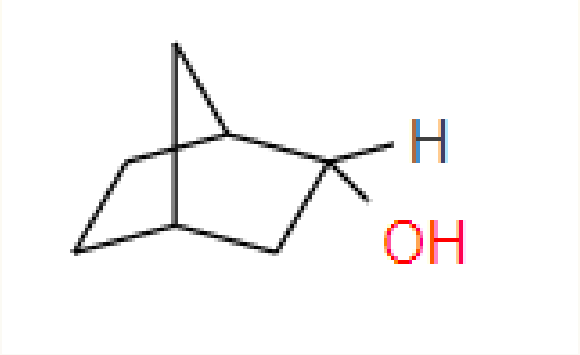
Bridged Ring Systems



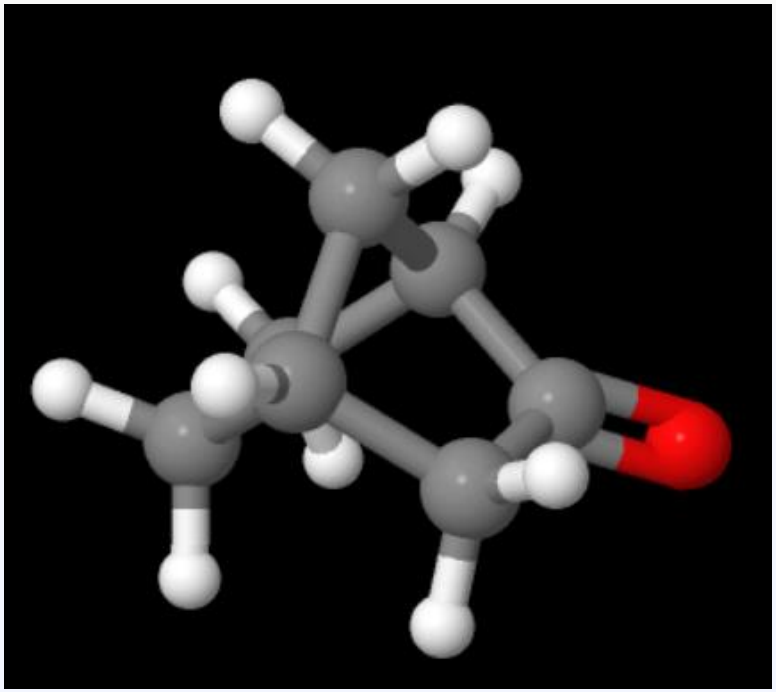
Norcamphor

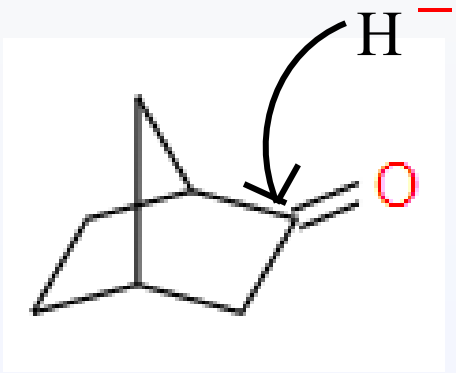
H_2, cat

or $LiAlH_4$



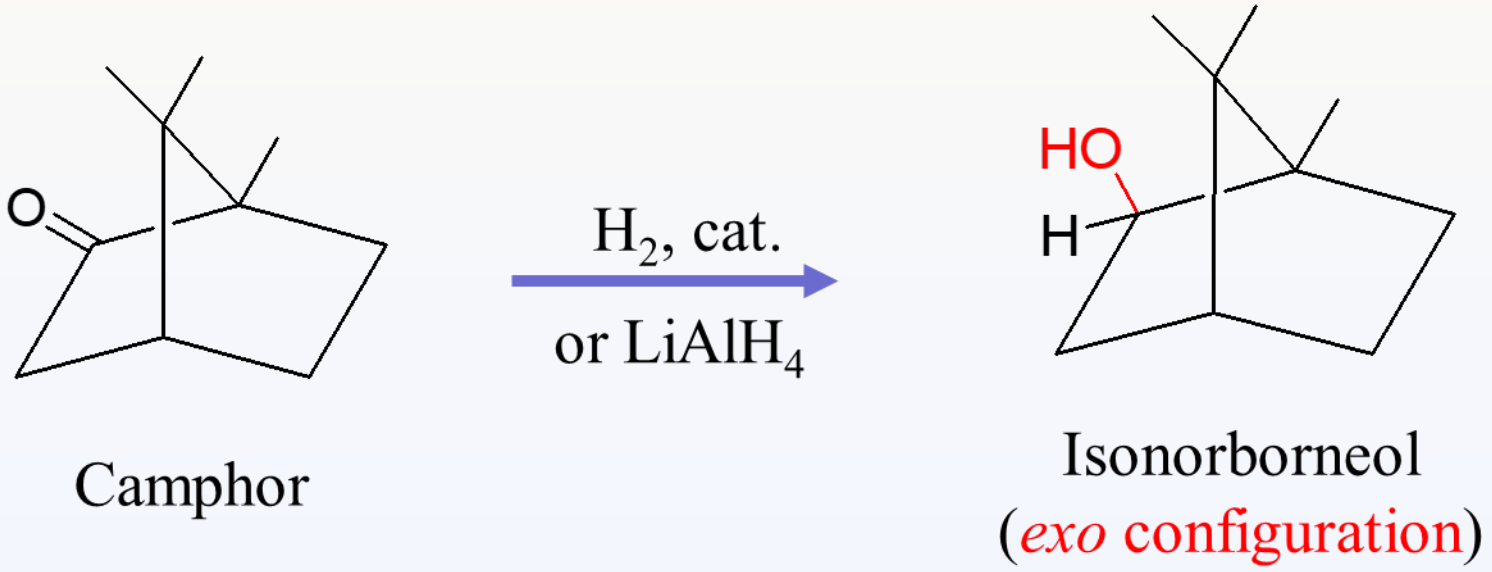
endo-Norborneol





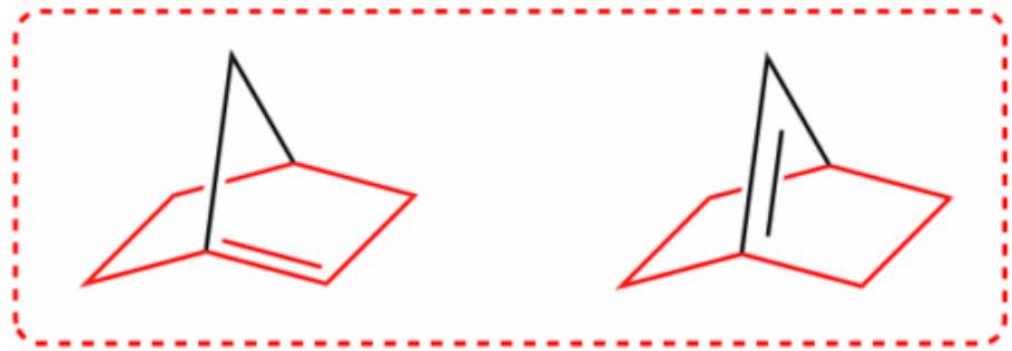
Bridged Ring Systems

- In the camphane system, the molecule is strongly screened by the *gem*-dimethyl group at C_7.
- This system is generally approached by reagents from the endo side.

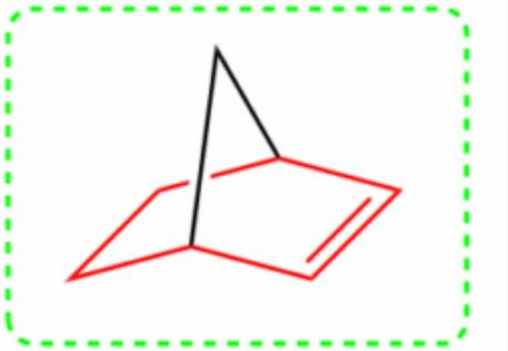


The Bredt Rule

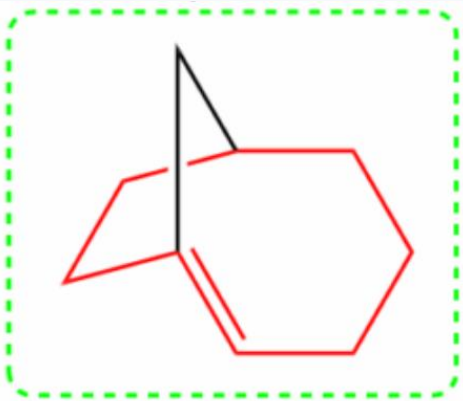
- The rule states that it is not possible for double bonds to be formed on the bridgehead carbon atoms of a bicyclic system if it involves a trans π bond being incorporated in a ring comprised of fewer than eight atoms.



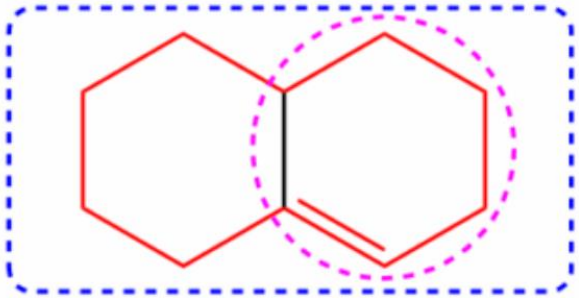
Not possible according to Bredt's Rule
- trans pi double bonded carbon is in a ring with less than eight carbon atoms



Possible
- trans pi double bonded carbon is not the bridgehead carbon



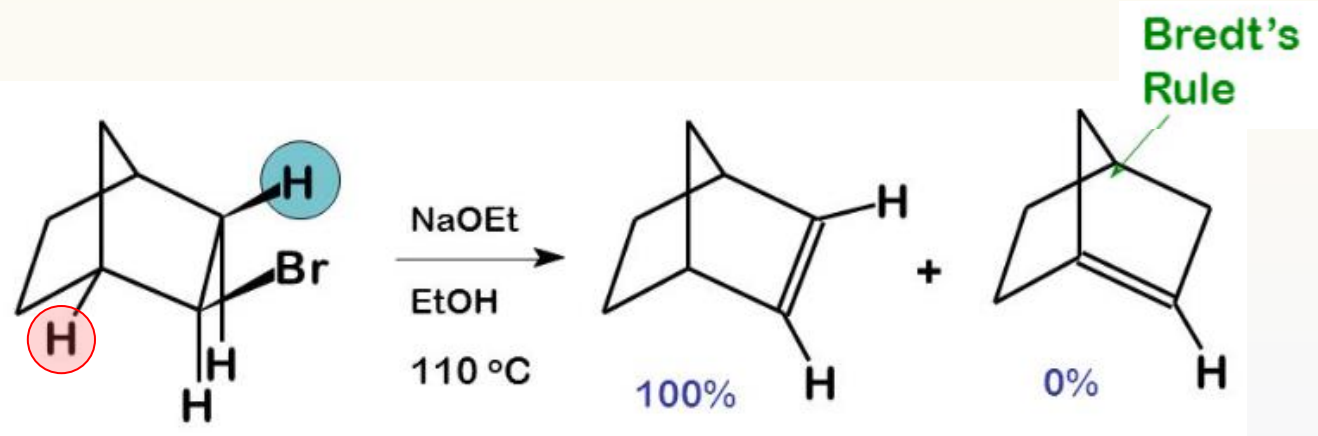
Possible
- trans pi double bonded carbon is in a ring with eight carbon atoms

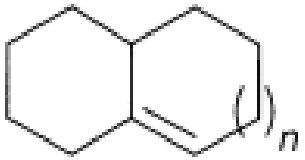
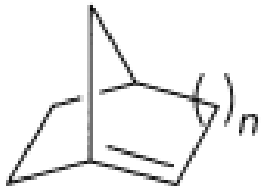


Possible
- cis in 6 carbon atom ring
- trans in 10 carbon atom ring

The Bredt Rule

- During the β-eliminations in bicyclic systems, the bridgehead atoms are not involved, provided the ring bearing the incipient trans double bond has at least eight atoms.



Fused	Bridged
	
Double bond to the bridgehead is stable (independent of *n*).	Double bond to the bridgehead is highly strained when $n < 3$.